Doubly Stochastic Primal–Dual Hybrid Gradient Method for Saddle-Point Problems with Nonlinear Coupling Terms

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Abstract

We consider convex-concave saddle-point problems with nonsmooth block-separable primal and dual terms and a smooth nonlinear block-separable coupling. Motivated by large-scale applications where a full update on either side can be computationally expensive, we propose a nonlinear doubly stochastic primal-dual hybrid gradient method (NL-DSPDHG) that performs randomized block updates on both primal and dual variables. For the general convex setting, we establish an $\mathcal{O}(1/K)$ ergodic convergence rate for the expected restricted primal–dual gap. A central difficulty is the uniform control of two comparison-point-indexed nonlinear martingale fields; this is achieved by combining martingale orthogonality with Sobolev embedding on the compact primal and dual domains. We further analyze a restarted variant of NL-DSPDHG under a quadratic growth condition in terms of the smoothed primal-dual gap. Under this regularity assumption, we prove linear convergence of the restarted outer iterates.

1 Introduction

Convex–concave saddle-point problems provide a unified formulation for constrained convex optimization, variational inequalities, inverse problems, and minimax models. We consider problems of the form

$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}}; L(x,y) := g(x) + \Phi(x,y) - f(y), \tag{1.1}$$

where g and f are proper lower semicontinuous convex functions and the coupling function Φ is convex in x, concave in y, and smooth. Our focus is the block-structured setting

$$g(x) = \sum_{j=1}^{m} g_j(x_j), \qquad f(y) = \sum_{i=1}^{n} f_i(y_i), \qquad \Phi(x,y) = \sum_{i=1}^{n}\sum_{j=1}^{m} \Phi_{ij}(x_j, y_i). \tag{1.2}$$

When the numbers of primal and dual blocks are large, a full iteration may require evaluating all block partial gradients and proximal mappings even when each individual block interaction is inexpensive. This motivates methods that exploit separability on both sides and update only randomly selected subsets of primal and dual blocks.

Related work. For smooth convex–concave saddle-point problems and monotone variational inequalities, the extragradient and Mirror–Prox viewpoints provide the classical $\mathcal{O}(1/K)$ ergodic complexity under Lipschitz continuity of the associated monotone operator [10]; stochastic Mirror–Prox extends this framework to noisy operator evaluations [9]. For composite problems with a bilinear coupling, the primal–dual hybrid gradient (PDHG) method of Chambolle and Pock [4, 5] has become a standard splitting method because it treats the nonsmooth primal and dual terms by separate proximal steps and accesses the coupling through inexpensive linear operations.

Randomization can substantially reduce the cost of a PDHG iteration in block-separable problems. The stochastic PDHG method of Chambolle, Ehrhardt, Richt'arik, and Sch"onlieb

updates randomly sampled dual blocks and permits general dual-side sampling rules [3]. Its subsequent convergence theory includes almost-sure convergence, an expected $\mathcal{O}(1/K)$ ergodic primal–dual gap rate in the general convex case, and linear convergence under metric subregularity [2]. More recently, randomized block-coordinate primal–dual schemes that sample both primal and dual variables have been developed for large-scale deterministic and stochastic saddle-point problems [8].

The replacement of a bilinear coupling by a genuinely nonlinear one changes both the algorithmic design and the convergence analysis. Nonlinear extensions of PDHG-type splitting were studied by Valkonen in the context of nonlinear operators [11]. For general smooth convex–concave couplings, Hamedani and Aybat proposed a primal–dual algorithm with a momentum correction and line search and proved an $\mathcal{O}(1/K)$ ergodic rate in the convex setting [7]. Zhu, Liu, and Tran-Dinh developed a single-loop primal–dual framework for a class of nonsmooth nonlinear convex–concave minimax problems and established $\mathcal{O}(1/K)$ guarantees under general convexity [12]. These works demonstrate that nonlinear coupling and block randomization can each be handled effectively. The present paper addresses their combination from a PDHG perspective and focuses on a different set of analytical questions: two successive primal and dual sampling stages, a restricted-gap guarantee for the actual stochastic iterates, local rather than global smoothness obtained through compact-domain proximal steps, and restart under a smoothed-gap quadratic growth condition.

The nonlinear doubly stochastic setting presents several difficulties that are absent from standard bilinear SPDHG. First, the primal and dual samples are revealed at different half-steps. Consequently, the sampling-corrected gradient estimators are centered with respect to different conditional sigma-fields, and the proof must preserve the corresponding filtration structure throughout the iteration. Second, nonlinear Taylor remainders and temporal gradient variations couple the current update with previous random increments. A Lyapunov function based only on weighted distances is therefore insufficient; additional nonlinear correction terms are needed to recover descent. Third, after summing the realized one-step inequalities and taking the supremum over comparison points, one encounters the random fields

$$y \longmapsto \sum_{k=0}^{K-1} \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle, \qquad x \longmapsto \sum_{k=0}^{K-1} \left\langle \nabla_y \Phi(x, y^k), v^k \right\rangle, \tag{1.3}$$

where u^k and v^k are centered block-sampling errors. Pointwise martingale cancellation does not control the supremum, because the comparison point is selected after the whole trajectory has been realized. We overcome this difficulty by estimating suitable Sobolev norms of the fields in (1.3) and then applying Sobolev embedding on the compact comparison domains [1].

The restart analysis creates a further obstacle. The Sobolev maximal estimate is homogeneous of degree one in the martingale increments and consequently produces a square-root dependence on their accumulated conditional variance. Such a bound is sufficient for an $\mathcal{O}(1/K)$ ergodic theorem, but it does not directly close a contraction recurrence in the squared distance to the solution set. Motivated by the smoothed-gap error-bound framework for restarted averaged PDHG [6], we retain the negative quadratic smoothing penalty inside the supremum. This penalty absorbs the Lipschitz variation of the two nonlinear martingale fields and reduces their contribution to the variance level, which is controlled by the initial squared distance of the epoch.

We propose NL-DSPDHG, a nonlinear doubly stochastic PDHG method for (1.1)–(1.2). At each iteration, a random subset of primal blocks is updated at the first half-step and a random subset of dual blocks is updated at the second half-step. Sampling corrections restore conditional unbiasedness, while a reflected primal-gradient term provides the temporal correction required by the nonlinear Lyapunov argument. Only the marginal inclusion probabilities are used; independence among block indicators within the same sampled subset is not required. When both sampling probabilities equal one, the method reduces to a deterministic nonlinear

primal–dual scheme, and setting only one probability equal to one gives a one-sided stochastic specialization.

A key modeling feature is that every block proximal step is performed over a prescribed compact box. Boundedness is therefore enforced by the algorithm rather than postulated as an a priori property of the random iterates. The construction also identifies a compact region containing all actual iterates, virtual full updates, and extrapolated points. Hence the Lipschitz and higher-order regularity assumptions on Φ need only hold on a neighborhood of this explicitly defined region. In particular, the higher differentiability used in the Sobolev argument is isolated from the first- and second-order assumptions required for stability and restart.

Contributions. The main contributions of this paper are summarized below.

(i) **Nonlinear doubly stochastic PDHG.** We derive a PDHG-type method that independently randomizes the primal and dual block updates at two successive half-steps. The method combines sampling-corrected nonlinear partial gradients with a reflected primal-gradient correction and retains fully block-separable proximal subproblems. ‟

(ii) **Stability and expected $\mathcal{O}(1/K)$ restricted-gap convergence.** We establish a realized one-step primal–dual gap inequality and construct a nonlinear Lyapunov function that absorbs the coupling variations and reflected-gradient terms under explicit step-size and sampling conditions. The resulting descent estimate yields stability of the stochastic trajectory, square-summability of its increments, and square-summability of all sampling and nonlinear-gradient errors. For the ergodic average of the actual stochastic iterates, we prove an expected restricted primal–dual gap bound of order $\mathcal{O}(1/K)$. The two nonlinear martingale fields in (1.3) are controlled uniformly through martingale orthogonality and Sobolev embedding, whereas the remaining linear martingale terms are treated directly at the conditional-variance level.

(iii) **Linear convergence under smoothed-gap quadratic growth.** We analyze a fixed-period restarted ergodic variant under a quadratic growth condition for the smoothed primal–dual gap. By keeping the negative quadratic penalty inside the defining supremum, we absorb the comparison-point variation of the nonlinear martingale fields and obtain a one-epoch estimate at the squared-distance level. This leads to a closed two-step outer recurrence and geometric convergence of both the expected smoothed gap and the expected squared distance to the saddle-point set. With the explicit epoch-length choice in the main theorem, the outer iterates satisfy a bound of order 5^{-t}. ‟

The analysis is presented for scalar primal and dual step sizes and uniform marginal probabilities in order to isolate the nonlinear and stochastic mechanisms. Extensions to block-dependent metrics and nonuniform sampling would require replacing the elementary conditional variance identities by suitable blockwise expected-separable-overapproximation bounds.

The remainder of the paper is organized as follows. Section 2 introduces the compact-domain formulation, the local regularity assumptions, and the restricted primal–dual gap. Section 3 presents NL-DSPDHG and establishes the conditional unbiasedness identities associated with the two half-step filtrations. The subsequent sections develop the virtual-update estimates, nonlinear Lyapunov descent, realized one-step inequality, and summability bounds for the stochastic errors, culminating in the expected $\mathcal{O}(1/K)$ restricted-gap theorem. Finally, we introduce the smoothed gap and the restarted scheme, prove the one-epoch estimate, and establish linear convergence under smoothed-gap quadratic growth.

2 Problem formulation and compact-domain construction

Let $m, n \in \mathbb{N}$, and let

$$x = (x_1, \dots, x_m) \in \mathbb{R}^M, \qquad y = (y_1, \dots, y_n) \in \mathbb{R}^N,$$

where $x_j \in \mathbb{R}^{M_j}$, $y_i \in \mathbb{R}^{N_i}$, $M = \sum_{j=1}^m M_j$, and $N = \sum_{i=1}^n N_i$. We study

$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} L(x,y), \qquad L(x,y) = g(x) + \Phi(x,y) - f(y), \tag{2.1}$$

with

$$g(x) = \sum_{j=1}^m g_j(x_j), \qquad f(y) = \sum_{i=1}^n f_i(y_i), \qquad \Phi(x,y) = \sum_{i=1}^n \sum_{j=1}^m \Phi_{ij}(x_j, y_i). \tag{2.2}$$

For every j and i, fix nonempty compact cubes

$$\mathcal{X}_j = [\underline{x}_j, \overline{x}_j]^{M_j}, \qquad \mathcal{Y}_i = [\underline{y}_i, \overline{y}_i]^{N_i},$$

and set

$$\mathcal{X} = \prod_{j=1}^m \mathcal{X}_j, \qquad \mathcal{Y} = \prod_{i=1}^n \mathcal{Y}_i, \qquad \mathcal{Z} = \mathcal{X} \times \mathcal{Y}.$$

The constrained proximal maps are

$$\operatorname{prox}_{\tau g_j}^{\mathcal{X}_j}(a) := \arg\min_{u\in\mathcal{X}_j} \left\{ g_j(u) + \frac{1}{2\tau} \|u-a\|^2 \right\}, \tag{2.3}$$

$$\operatorname{prox}_{\sigma f_i}^{\mathcal{Y}_i}(b) := \arg\min_{v\in\mathcal{Y}_i} \left\{ f_i(v) + \frac{1}{2\sigma} \|v-b\|^2 \right\}. \tag{2.4}$$

Equivalently, these are the ordinary proximal maps of $g_j + \delta_{\mathcal{X}_j}$ and $f_i + \delta_{\mathcal{Y}_i}$. Every actual update and every virtual full update therefore belongs to the corresponding compact cube automatically.

Let $p, q \in (0,1]$ be the uniform primal and dual marginal probabilities. The extrapolated iterates have the form

$$\bar{x}^{k+1} = x^k + \frac{1}{p}(x^{k+1} - x^k), \qquad \bar{y}^{k+1} = y^k + \frac{1}{q}(y^{k+1} - y^k).$$

Define the enlarged compact sets

$$\mathcal{X}_p^{\mathrm{ext}} := \mathcal{X} + \frac{1}{p}(\mathcal{X} - \mathcal{X}), \qquad \mathcal{Y}_q^{\mathrm{ext}} := \mathcal{Y} + \frac{1}{q}(\mathcal{Y} - \mathcal{Y}), \qquad \mathcal{K} := \mathcal{X}_p^{\mathrm{ext}} \times \mathcal{Y}_q^{\mathrm{ext}}. \tag{2.5}$$

All values and derivatives of Φ used by the algorithm or the proof are evaluated in $\mathcal{K}$.

For $a > 0$, let $\|u\|_a^2 = a\|u\|^2$. We use

$$\|z\|_V^2 := \|x\|_{\tau^{-1}p^{-1}}^2 + \|y\|_{\sigma^{-1}q^{-1}}^2, \qquad z = (x,y). \tag{2.6}$$

Assumption 1 (Convexity, compact-domain regularity, and existence)**.** *For every j and i, g_j and f_i are proper, lower semicontinuous, convex, and finite on $\mathcal{X}_j$ and $\mathcal{Y}_i$, respectively. Each Φ_{ij} is convex in its first variable, concave in its second variable, and C^{r_0} on an open neighborhood of $\mathcal{K}$, where*

$$r_0 > \frac{\max\{M,N\}}{2} + 1.$$

The restricted saddle-point set

$$\mathcal{Z}^* := \{(x^*, y^*) \in \mathcal{Z} : L(x^*, y) \le L(x^*, y^*) \le L(x, y^*) \quad \forall (x,y) \in \mathcal{Z}\} \tag{2.7}$$

is nonempty.

The derivative order in Assumption 1 is needed only for the Sobolev maximal estimate in the $O(1/K)$ restricted-gap theorem. The stability and restart arguments use only first- and second-order bounds on $\mathcal{K}$.

Assumption 2 (Local Lipschitz and Hessian constants)**.** *There exist nonnegative constants*

$$L^{xx}, \quad L^{xy}, \quad L^{yx}, \quad L^{yy}, \quad B$$

such that, for all $(x_1, y_1), (x_2, y_2) \in \mathcal{K}$,

$$\|\nabla_x \Phi(x_1, y_1) - \nabla_x \Phi(x_2, y_2)\| \le L^{xx} \|x_1 - x_2\| + L^{xy} \|y_1 - y_2\|, \tag{2.8}$$

$$\|\nabla_y \Phi(x_1, y_1) - \nabla_y \Phi(x_2, y_2)\| \le L^{yx} \|x_1 - x_2\| + L^{yy} \|y_1 - y_2\|, \tag{2.9}$$

$$\left\|\nabla^2 \Phi(x, y)\right\| \le B. \tag{2.10}$$

For the endpoint estimates in the restart analysis, define

$$\ell_{xy} := \sqrt{\tau\sigma pq} \sup_{(x,y)\in\mathcal{K}} \left\|\nabla^2 \Phi(x, y)\right\|. \tag{2.11}$$

Then the mixed four-point difference is bounded by ℓ_{xy} *times the product of the corresponding* $\tau^{-1}p^{-1}$*- and* $\sigma^{-1}q^{-1}$*-weighted norms.*

The primal–dual gap kernel is

$$H(\bar{z}, z) := H(\bar{x}, \bar{y}; x, y) := L(\bar{x}, y) - L(x, \bar{y}). \tag{2.12}$$

For $z^* \in \mathcal{Z}^*$, $H(\bar{z}, z^*) \ge 0$ for every $\bar{z} \in \mathcal{Z}$. For a nonempty compact $\mathcal{B} \subseteq \mathcal{Z}$, define the restricted gap

$$G_{\mathcal{B}}(\bar{z}) := \sup_{z\in\mathcal{B}} H(\bar{z}, z). \tag{2.13}$$

3 The nonlinear doubly stochastic PDHG method

Let $S^{k+1} \subseteq [m]$ and $T^{k+1} \subseteq [n]$ be random subsets satisfying

$$\mathbb{P}(j \in S^{k+1}) = p, \qquad \mathbb{P}(i \in T^{k+1}) = q. \tag{3.1}$$

The sets $S^1, T^1, S^2, T^2, \ldots$ are independent. Independence among indicators within a fixed subset is not required; only the common marginals and the block-separable representation (2.2) are used in the conditional expectation identities.

Initialize

$$(x^{-2}, y^{-2}) = (x^{-1}, y^{-1}) = (x^0, y^0) \in \mathcal{Z}, \qquad w^{-1} = w^0 = \nabla_x \Phi(x^0, y^0), \qquad s^0 = w^0. \tag{3.2}$$

Define

$$\bar{x}^{k+1} = x^k + p^{-1}(x^{k+1} - x^k), \tag{3.3}$$

$$\bar{y}^{k+1} = y^k + q^{-1}(y^{k+1} - y^k), \tag{3.4}$$

$$l^k = \nabla_y \Phi(x^k, y^k) + \frac{1}{p}\big(\nabla_y \Phi(x^{k+1}, y^k) - \nabla_y \Phi(x^k, y^k)\big), \tag{3.5}$$

$$w^{k+1} = \nabla_x \Phi(\bar{x}^{k+1}, y^k) + \frac{1}{q}\big(\nabla_x \Phi(\bar{x}^{k+1}, y^{k+1}) - \nabla_x \Phi(\bar{x}^{k+1}, y^k)\big), \tag{3.6}$$

$$s^{k+1} = 2w^{k+1} - w^k \tag{3.7}$$

Let

$$\mathcal{F}_k := \sigma(S^1, T^1, \ldots, S^k, T^k), \qquad \mathcal{F}_{k+\frac{1}{2}} := \sigma(\mathcal{F}_k, S^{k+1}). \tag{3.8}$$

Algorithm 1 Nonlinear doubly stochastic PDHG (NL-DSPDHG)

Require: $x^0 \in \mathcal{X}$, $y^0 \in \mathcal{Y}$, $\tau, \sigma > 0$, $p, q \in (0, 1]$

initialize as in (3.2)

for $k = 0, 1, 2, \ldots$ **do**

  Draw $S^{k+1} \subseteq [m]$

  Compute s_j^k as in (3.7)

  Update $x_j^{k+1} = \begin{cases} \operatorname{prox}_{\tau g_j}^{\mathcal{X}_j}(x_j^k - \tau s_j^k), & j \in S^{k+1}, \\ x_j^k, & j \notin S^{k+1} \end{cases}$

  Draw $T^{k+1} \subseteq [n]$

  Compute l_i^k as in (3.5)

  Update $y_i^{k+1} = \begin{cases} \operatorname{prox}_{\sigma f_i}^{\mathcal{Y}_i}(y_i^k + \sigma l_i^k), & i \in T^{k+1}, \\ y_i^k, & i \notin T^{k+1} \end{cases}$

end for

Define the virtual full proximal updates

$$\hat{x}_j^{k+1} := \operatorname{prox}_{\tau g_j}^{\mathcal{X}_j}(x_j^k - \tau s_j^k), \qquad j = 1, \ldots, m, \tag{3.9}$$

$$\hat{y}_i^{k+1} := \operatorname{prox}_{\sigma f_i}^{\mathcal{Y}_i}(y_i^k + \sigma l_i^k), \qquad i = 1, \ldots, n. \tag{3.10}$$

Thus $x_j^{k+1} = \hat{x}_j^{k+1}$ on S^{k+1} and is equal to x_j^k otherwise; the analogous statement holds for y^{k+1}.

Lemma 1 (Conditional unbiasedness)**.** *For every $k \geq 0$,*

$$\mathbb{E}[\bar{x}^{k+1} \mid \mathcal{F}_k] = \hat{x}^{k+1}, \tag{3.11}$$

$$\mathbb{E}[l^k \mid \mathcal{F}_k] = \nabla_y \Phi(\hat{x}^{k+1}, y^k), \tag{3.12}$$

$$\mathbb{E}[\bar{y}^{k+1} \mid \mathcal{F}_{k+\frac{1}{2}}] = \hat{y}^{k+1}, \tag{3.13}$$

$$\mathbb{E}[w^{k+1} \mid \mathcal{F}_{k+\frac{1}{2}}] = \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}). \tag{3.14}$$

Proof. For each primal block, let $d_j^k = \hat{x}_j^{k+1} - x_j^k$. Then $x_j^{k+1} - x_j^k = \mathbf{1}_{\{j \in S^{k+1}\}} d_j^k$, and hence

$$\mathbb{E}[x_j^{k+1} - x_j^k \mid \mathcal{F}_k] = p d_j^k.$$

Consequently,

$$\mathbb{E}[\bar{x}_j^{k+1} \mid \mathcal{F}_k] = x_j^k + p^{-1}\mathbb{E}[x_j^{k+1} - x_j^k \mid \mathcal{F}_k] = x_j^k + d_j^k = \hat{x}_j^{k+1}.$$

Because Φ is block separable,

$$\nabla_y \Phi(x, y) = \sum_{j=1}^{m} \nabla_y \Phi_j(x_j, y), \qquad \Phi_j(x_j, y) := \sum_{i=1}^{n} \Phi_{ij}(x_j, y_i).$$

Therefore, the random correction in l^k is a sum of selected block increments, and taking the conditional expectation replaces it by the full increment from x^k to $\hat{x}^{k+1}$. This proves (3.12). The dual identities follow in exactly the same manner after conditioning on $\mathcal{F}_{k+\frac{1}{2}}$. □

Define the centered errors

$$u^k := \hat{x}^{k+1} - \bar{x}^{k+1}, \qquad v^k := \hat{y}^{k+1} - \bar{y}^{k+1}, \tag{3.15}$$

$$e_y^k := \nabla_y \Phi(\hat{x}^{k+1}, y^k) - l^k, \qquad e_x^{k+1} := w^{k+1} - \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}). \tag{3.16}$$

Then

$$\mathbb{E}[u^k \mid \mathcal{F}_k] = \mathbb{E}[e_y^k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[v^k \mid \mathcal{F}_{k+\frac{1}{2}}] = \mathbb{E}[e_x^{k+1} \mid \mathcal{F}_{k+\frac{1}{2}}] = 0. \tag{3.17}$$

4 Virtual one-step estimates and expectation replacement

For $z = (x, y) \in \mathcal{Z}$, define

$$\widehat{Q}_k(z) := \frac{1}{2}\left\|x^k - x\right\|_{\tau^{-1}}^2 + \frac{1}{2}\left\|y^k - y\right\|_{\sigma^{-1}}^2 - \frac{1}{2}\left\|\hat{x}^{k+1} - x\right\|_{\tau^{-1}}^2 - \frac{1}{2}\left\|\hat{y}^{k+1} - y\right\|_{\sigma^{-1}}^2 - \frac{1}{2}\left\|\hat{x}^{k+1} - x^k\right\|_{\tau^{-1}}^2 - \frac{1}{2}\left\|\hat{y}^{k+1} - y^k\right\|_{\sigma^{-1}}^2. \tag{4.1}$$

Lemma 2 (Proximal three-point gap inequality)**.** *For every $k \geq 0$ and $z = (x, y) \in \mathcal{Z}$,*

$$H(\hat{x}^{k+1}, \hat{y}^{k+1}; x, y) \leq \left\langle s^k, x - \hat{x}^{k+1}\right\rangle + \left\langle l^k, \hat{y}^{k+1} - y\right\rangle + \Phi(\hat{x}^{k+1}, y) - \Phi(x, \hat{y}^{k+1}) + \widehat{Q}_k(z). \tag{4.2}$$

Proof. The optimality condition for (3.9) is

$$0 \in \partial(g + \delta_{\mathcal{X}})(\hat{x}^{k+1}) + s^k + \tau^{-1}(\hat{x}^{k+1} - x^k).$$

Testing the subgradient inequality at $x \in \mathcal{X}$ gives

$$\begin{aligned} g(\hat{x}^{k+1}) - g(x) &\leq \left\langle s^k, x - \hat{x}^{k+1}\right\rangle + \left\langle \tau^{-1}(\hat{x}^{k+1} - x^k), x - \hat{x}^{k+1}\right\rangle \\ &= \left\langle s^k, x - \hat{x}^{k+1}\right\rangle + \frac{1}{2}\left\|x^k - x\right\|_{\tau^{-1}}^2 - \frac{1}{2}\left\|\hat{x}^{k+1} - x\right\|_{\tau^{-1}}^2 - \frac{1}{2}\left\|\hat{x}^{k+1} - x^k\right\|_{\tau^{-1}}^2. \end{aligned}$$

The dual optimality condition gives

$$f(\hat{y}^{k+1}) - f(y) \leq \left\langle l^k, \hat{y}^{k+1} - y\right\rangle + \frac{1}{2}\left\|y^k - y\right\|_{\sigma^{-1}}^2 - \frac{1}{2}\left\|\hat{y}^{k+1} - y\right\|_{\sigma^{-1}}^2 - \frac{1}{2}\left\|\hat{y}^{k+1} - y^k\right\|_{\sigma^{-1}}^2.$$

Adding these inequalities and the coupling difference proves (4.2). □

By concavity in y and convexity in x,

$$\Phi(\hat{x}^{k+1}, y) \leq \Phi(\hat{x}^{k+1}, y^k) + \left\langle \nabla_y \Phi(\hat{x}^{k+1}, y^k), y - y^k\right\rangle, \tag{4.3}$$

$$\Phi(x, \hat{y}^{k+1}) \geq \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}) + \left\langle \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}), x - \bar{x}^{k+1}\right\rangle. \tag{4.4}$$

Define

$$\begin{aligned} \widehat{\Gamma}_k(z) := &\left\langle s^k, x - \hat{x}^{k+1}\right\rangle + \left\langle l^k, \hat{y}^{k+1} - y\right\rangle + \widehat{Q}_k(z) \\ &+ \Phi(\hat{x}^{k+1}, y^k) - \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}) \\ &+ \left\langle \nabla_y \Phi(\hat{x}^{k+1}, y^k), y - y^k\right\rangle - \left\langle \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}), x - \bar{x}^{k+1}\right\rangle. \end{aligned} \tag{4.5}$$

Then Lemma 2 and (4.3)–(4.4) imply

$$H(\hat{x}^{k+1}, \hat{y}^{k+1}; x, y) \leq \widehat{\Gamma}_k(z). \tag{4.6}$$

Define

$$\begin{aligned} \Lambda_k := &\Phi(x^k, y^k) + \frac{1}{p}\big(\Phi(x^{k+1}, y^k) - \Phi(x^k, y^k)\big) - \Phi(\bar{x}^{k+1}, y^k) \\ &- \frac{1}{q}\big(\Phi(\bar{x}^{k+1}, y^{k+1}) - \Phi(\bar{x}^{k+1}, y^k)\big) + \frac{1}{q}\left\langle l^k, y^{k+1} - y^k\right\rangle, \end{aligned} \tag{4.7}$$

$$\begin{aligned} Q_k(z) := &\frac{1}{2}\left\|x^k - x\right\|_{\tau^{-1}p^{-1}}^2 + \frac{1}{2}\left\|y^k - y\right\|_{\sigma^{-1}q^{-1}}^2 - \frac{1}{2}\left\|x^{k+1} - x\right\|_{\tau^{-1}p^{-1}}^2 - \frac{1}{2}\left\|y^{k+1} - y\right\|_{\sigma^{-1}q^{-1}}^2 \\ &- \frac{1}{2}\left\|x^{k+1} - x^k\right\|_{\tau^{-1}p^{-1}}^2 - \frac{1}{2}\left\|y^{k+1} - y^k\right\|_{\sigma^{-1}q^{-1}}^2, \end{aligned} \tag{4.8}$$

$$\Gamma_k(z) := \left\langle w^{k+1} - s^k, \bar{x}^{k+1} - x\right\rangle + \Lambda_k + Q_k(z). \tag{4.9}$$

Lemma 3 (Expectation replacement)**.** *For every deterministic or $\mathcal{F}_k$-measurable $\mathcal{Z}$-valued comparison point z,*

$$\mathbb{E}[\widehat{\Gamma}_k(z)] = \mathbb{E}[\Gamma_k(z)]. \tag{4.10}$$

Proof. We first compare $Q_k(z)$ and $\widehat{Q}_k(z)$. For the primal part, the blockwise update rule implies

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{2}\left\|x^k - x\right\|^2_{\tau^{-1}p^{-1}} - \frac{1}{2}\left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} - \frac{1}{2}\left\|x^{k+1} - x\right\|^2_{\tau^{-1}p^{-1}} \,\middle|\, \mathcal{F}_k\right] \\
&\quad = \frac{1}{2}\left\|x^k - x\right\|^2_{\tau^{-1}} - \frac{1}{2}\left\|\hat{x}^{k+1} - x^k\right\|^2_{\tau^{-1}} - \frac{1}{2}\left\|\hat{x}^{k+1} - x\right\|^2_{\tau^{-1}}.
\end{aligned} \tag{4.11}$$

Indeed, for every $\mathcal{F}_k$-measurable random vector x and every $j \in [m]$,

$$\mathbb{E}\left[\left\|x_j^{k+1} - x_j\right\|^2 \mid \mathcal{F}_k\right] = p\left\|\hat{x}_j^{k+1} - x_j\right\|^2 + (1-p)\left\|x_j^k - x_j\right\|^2.$$

The same argument, conditionally on $\mathcal{F}_{k+\frac{1}{2}}$, gives

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{2}\left\|y^k - y\right\|^2_{\sigma^{-1}q^{-1}} - \frac{1}{2}\left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}} - \frac{1}{2}\left\|y^{k+1} - y\right\|^2_{\sigma^{-1}q^{-1}} \,\middle|\, \mathcal{F}_{k+\frac{1}{2}}\right] \\
&\quad = \frac{1}{2}\left\|y^k - y\right\|^2_{\sigma^{-1}} - \frac{1}{2}\left\|\hat{y}^{k+1} - y^k\right\|^2_{\sigma^{-1}} - \frac{1}{2}\left\|\hat{y}^{k+1} - y\right\|^2_{\sigma^{-1}}.
\end{aligned} \tag{4.12}$$

Taking iterated expectations in (4.11) and (4.12) yields

$$\mathbb{E}[\widehat{Q}_k(z)] = \mathbb{E}[Q_k(z)]. \tag{4.13}$$

Next, block separability implies

$$\begin{aligned}
\mathbb{E}\left[\Phi(x^k, y^k) + \frac{1}{p}\big(\Phi(x^{k+1}, y^k) - \Phi(x^k, y^k)\big) \,\middle|\, \mathcal{F}_k\right] &= \Phi(\hat{x}^{k+1}, y^k), \\
\mathbb{E}\left[\Phi(\bar{x}^{k+1}, y^k) + \frac{1}{q}\big(\Phi(\bar{x}^{k+1}, y^{k+1}) - \Phi(\bar{x}^{k+1}, y^k)\big) \,\middle|\, \mathcal{F}_{k+\frac{1}{2}}\right] &= \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}).
\end{aligned}$$

Lemma 1 similarly replaces l^k and w^{k+1} by their conditional mean gradients. Taking the two successive conditional expectations and then the total expectation gives (4.10). □

5 Noise decomposition and nonlinear Lyapunov descent

The difference between the virtual majorant $\widehat{\Gamma}_k$ and the realized quantity Γ_k contains four vector martingale differences and one scalar centered remainder. The explicit decomposition is useful both for the restricted-gap proof and for the restarted analysis.

Let

$$\widetilde{\Lambda}_k := \Phi(\hat{x}^{k+1}, y^k) - \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}) + \left\langle l^k, \hat{y}^{k+1} - y^k \right\rangle, \tag{5.1}$$

and define the scalar quadratic discrepancy

$$\begin{aligned}
\delta_k := &\frac{1}{2}\left(\left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} - \left\|\hat{x}^{k+1} - x^k\right\|^2_{\tau^{-1}}\right) + \frac{1}{2}\left(\left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}} - \left\|\hat{y}^{k+1} - y^k\right\|^2_{\sigma^{-1}}\right) \\
&+ \frac{1}{2}\left(\left\|x^{k+1}\right\|^2_{\tau^{-1}p^{-1}} - \left\|x^k\right\|^2_{\tau^{-1}p^{-1}} + \left\|x^k\right\|^2_{\tau^{-1}} - \left\|\hat{x}^{k+1}\right\|^2_{\tau^{-1}}\right) \\
&+ \frac{1}{2}\left(\left\|y^{k+1}\right\|^2_{\sigma^{-1}q^{-1}} - \left\|y^k\right\|^2_{\sigma^{-1}q^{-1}} + \left\|y^k\right\|^2_{\sigma^{-1}} - \left\|\hat{y}^{k+1}\right\|^2_{\sigma^{-1}}\right).
\end{aligned} \tag{5.2}$$

Set

$$\begin{aligned}
\varepsilon_k := &-\left\langle s^k, u^k \right\rangle + \widetilde{\Lambda}_k - \Lambda_k + \delta_k \\
&+ \left\langle \bar{x}^{k+1}, \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}) - w^{k+1} \right\rangle + \left\langle y^k, l^k - \nabla_y \Phi(\hat{x}^{k+1}, y^k) \right\rangle.
\end{aligned} \tag{5.3}$$

Lemma 4 (Centered-noise decomposition)**.** *For every $z = (x, y) \in \mathcal{Z}$,*

$$\widehat{\Gamma}_k(z) - \Gamma_k(z) = \left\langle x, u^k \right\rangle_{\tau^{-1}} + \left\langle y, v^k \right\rangle_{\sigma^{-1}} + \left\langle x, e_x^{k+1} \right\rangle + \left\langle y, e_y^k \right\rangle + \varepsilon_k. \tag{5.4}$$

Moreover, $\mathbb{E}[\varepsilon_k] = 0$.

Proof. Using the identity

$$\|a - c\|_M^2 - \|b - c\|_M^2 = \|a\|_M^2 - \|b\|_M^2 + 2\langle c, b - a\rangle_M,$$

to expand $\widehat{Q}_k(z) - Q_k(z)$, we obtain

$$\widehat{Q}_k(z) - Q_k(z) = \left\langle x, u^k \right\rangle_{\tau^{-1}} + \left\langle y, v^k \right\rangle_{\sigma^{-1}} + \delta_k$$

Using

$$x - \hat{x}^{k+1} = x - \bar{x}^{k+1} - u^k, \qquad \hat{y}^{k+1} - y = \bar{y}^{k+1} - y + v^k,$$

adding and subtracting the two conditional mean gradients in (3.16) gives the terms involving e_x^{k+1} and e_y^k, and all remaining scalar terms are precisely (5.3). Taking expectations in (5.4), using Lemma 3, (3.17), and the tower property, shows that all four linear terms have zero expectation and hence $\mathbb{E}\varepsilon_k = 0$. □

We next record the Lyapunov construction. The following correction variables make the reflected-gradient terms telescope. For $k \geq 0$, define

$$A_k := \nabla_x \Phi(x^k, y^k) - \nabla_x \Phi(\bar{x}^k, y^k), \tag{5.5}$$

$$B_k := \nabla_x \Phi(\bar{x}^k, y^k) - \nabla_x \Phi(\bar{x}^k, y^{k-1}), \tag{5.6}$$

$$C_{k+1} := \nabla_x \Phi(\bar{x}^{k+1}, y^k) - \nabla_x \Phi(x^k, y^k). \tag{5.7}$$

With the convention that a quotient by L^{xx} or L^{xy} is zero whenever the corresponding Lipschitz constant and gradient difference both vanish, set

$$r_k^2 := \frac{p}{2L^{xx}} \|A_k\|^2 + \frac{1}{2L^{xy}} \left(\sqrt{\frac{\sigma\tau p}{q}} - \sqrt{\sigma\tau pq} \right) \|B_k\|_{\sigma^{-1}q^{-1}}^2, \tag{5.8}$$

$$d_{k+1}^2 := \frac{p}{2L^{xx}} \|C_{k+1}\|^2 + \frac{1}{2L^{xy}} \sqrt{\frac{\sigma\tau p}{q}} \|B_{k+1}\|_{\sigma^{-1}q^{-1}}^2 + r_k^2. \tag{5.9}$$

At initialization, $\bar{x}^0 = x^0$ and $\bar{y}^0 = y^0$, so $r_0 = 0$; we set $d_0 = 0$.

For $z = (x, y)$, define

$$\begin{aligned} \mathcal{V}_k(z) := {} & \frac{1}{2} \left\| x^k - x \right\|_{\tau^{-1}p^{-1}}^2 + \frac{1}{2} \left\| y^k - y \right\|_{\sigma^{-1}q^{-1}}^2 - \left\langle w^k - w^{k-1}, x^k - x \right\rangle \\ & + \frac{1}{p} d_k^2 + \left(\frac{2}{p} - 1 \right) r_k^2, \end{aligned} \tag{5.10}$$

$$\begin{aligned} \widetilde{\mathcal{V}}_k := {} & \frac{1}{2} \left\| x^{k+1} - x^k \right\|_{\tau^{-1}p^{-1}}^2 + \frac{1}{2} \left\| y^{k+1} - y^k \right\|_{\sigma^{-1}q^{-1}}^2 - \Lambda_k \\ & + \frac{1}{p} \left\langle w^k - w^{k-1}, x^{k+1} - x^k \right\rangle - \left(\frac{1}{p} - 1 \right) \left\langle w^{k+1} - w^k, x^{k+1} - x^k \right\rangle \\ & + \frac{1}{p} (d_k^2 - d_{k+1}^2) + \left(\frac{2}{p} - 1 \right) (r_k^2 - r_{k+1}^2). \end{aligned} \tag{5.11}$$

Lemma 5 (Exact Lyapunov identity)**.** *For every $k \geq 0$ and every $z \in \mathcal{Z}$,*

$$\Gamma_k(z) = \mathcal{V}_k(z) - \mathcal{V}_{k+1}(z) - \widetilde{\mathcal{V}}_k. \tag{5.12}$$

Proof. Because $s^k = 2w^k - w^{k-1}$,

$$\left\langle w^{k+1} - s^k, \bar{x}^{k+1} - x \right\rangle = \left\langle w^{k+1} - 2w^k + w^{k-1}, x^{k+1} - x \right\rangle + \left(\frac{1}{p} - 1\right) \left\langle w^{k+1} - 2w^k + w^{k-1}, x^{k+1} - x^k \right\rangle.$$

The first inner product can be rewritten as

$$\left\langle w^{k+1} - w^k, x^{k+1} - x \right\rangle - \left\langle w^k - w^{k-1}, x^k - x \right\rangle - \left\langle w^k - w^{k-1}, x^{k+1} - x^k \right\rangle.$$

Inserting this identity into (4.9), expanding $Q_k(z)$, and adding and subtracting the telescoping quantities $p^{-1}d_k^2$ and $(2p^{-1} - 1)r_k^2$ gives (5.12) term by term. No inequality is used in this step. □

The next assumption is a readable sufficient specialization of the coefficient conditions obtained by collecting the Young and Taylor estimates. It is local: all constants are the compact-region constants in Assumption 2.

Assumption 3 (Step-size and sampling conditions)**.** *There exists $\delta \in (0, 1)$ such that*

$$\frac{1-\delta}{2} \geq L^{xy}\sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2}\sqrt{\sigma\tau pq} + L^{xx}\tau, \tag{5.13}$$

$$\frac{1-\delta}{2} \geq \left(\frac{2}{p} - 1\right)\left[L^{xy}\sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2}\sqrt{\sigma\tau pq} + \frac{3 + (1-p)^2}{2}\tau L^{xx}\right] + \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} + \frac{\tau L^{xx}}{2} + \frac{\tau(B + L^{xx})}{2pq}, \tag{5.14}$$

$$\frac{1-\delta}{2} \geq \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} + \frac{\sigma B}{2} + \left(\frac{2}{p} - 1\right)\left[L^{xy}\sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2}\sqrt{\sigma\tau pq}\right]. \tag{5.15}$$

Lemma 6 (Coercivity and dissipation)**.** *Under Assumptions 1–3, for every $k \geq 0$ and $z = (x, y) \in \mathcal{Z}$,*

$$\mathcal{V}_k(z) \geq \frac{\delta}{2}\left\|x^k - x\right\|^2_{\tau^{-1}p^{-1}} + \frac{1}{2}\left\|y^k - y\right\|^2_{\sigma^{-1}q^{-1}}, \tag{5.16}$$

$$\widetilde{\mathcal{V}}_k \geq \frac{\delta}{2}\left(\left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} + \left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}}\right). \tag{5.17}$$

Proof. The proof is divided into four explicit estimates.

Step 1: an increment bound for $w^{k+1} - w^k$. By the definitions of w^{k+1}, A_k, B_k, and C_{k+1},

$$w^{k+1} = \nabla_x\Phi(\bar{x}^{k+1}, y^k) + \frac{1}{q}B_{k+1},$$
$$w^k = \nabla_x\Phi(\bar{x}^k, y^{k-1}) + \frac{1}{q}B_k,$$

and

$$\begin{aligned}\nabla_x\Phi(\bar{x}^{k+1}, y^k) - \nabla_x\Phi(\bar{x}^k, y^{k-1}) &= \left[\nabla_x\Phi(\bar{x}^{k+1}, y^k) - \nabla_x\Phi(x^k, y^k)\right] + \left[\nabla_x\Phi(x^k, y^k) - \nabla_x\Phi(\bar{x}^k, y^k)\right] \\ &\quad + \left[\nabla_x\Phi(\bar{x}^k, y^k) - \nabla_x\Phi(\bar{x}^k, y^{k-1})\right] = C_{k+1} + A_k + B_k.\end{aligned}$$

Consequently,

$$w^{k+1} - w^k = \frac{1}{q} B_{k+1} + \left(1 - \frac{1}{q}\right) B_k + C_{k+1} + A_k. \tag{5.18}$$

Let $h \in \mathbb{R}^M$. Since $\|h\| = \sqrt{\tau p}\, \|h\|_{\tau^{-1}p^{-1}}$ and $\|a\| = \sqrt{\sigma q}\, \|a\|_{\sigma^{-1}q^{-1}}$, Young's inequality gives

$$\frac{1}{q} |\langle B_{k+1}, h\rangle| \le \frac{L^{xy}}{2} \sqrt{\frac{\sigma\tau p}{q}}\, \|h\|^2_{\tau^{-1}p^{-1}} + \frac{1}{2L^{xy}} \sqrt{\frac{\sigma\tau p}{q}}\, \|B_{k+1}\|^2_{\sigma^{-1}q^{-1}}, \tag{5.19}$$

$$\begin{aligned}\left(\frac{1}{q} - 1\right) |\langle B_k, h\rangle| \le{}& \frac{L^{xy}}{2} \left(\sqrt{\frac{\sigma\tau p}{q}} - \sqrt{\sigma\tau pq}\right) \|h\|^2_{\tau^{-1}p^{-1}} \\ &+ \frac{1}{2L^{xy}} \left(\sqrt{\frac{\sigma\tau p}{q}} - \sqrt{\sigma\tau pq}\right) \|B_k\|^2_{\sigma^{-1}q^{-1}},\end{aligned} \tag{5.20}$$

$$|\langle C_{k+1}, h\rangle| \le \frac{\tau L^{xx}}{2} \|h\|^2_{\tau^{-1}p^{-1}} + \frac{p}{2L^{xx}} \|C_{k+1}\|^2, \tag{5.21}$$

$$|\langle A_k, h\rangle| \le \frac{\tau L^{xx}}{2} \|h\|^2_{\tau^{-1}p^{-1}} + \frac{p}{2L^{xx}} \|A_k\|^2. \tag{5.22}$$

The four gradient-square terms on the right-hand sides are exactly the terms in d^2_{k+1}. Hence

$$\left|\left\langle w^{k+1} - w^k, h\right\rangle\right| \le A_{xy} \|h\|^2_{\tau^{-1}p^{-1}} + d^2_{k+1}, \tag{5.23}$$

where

$$A_{xy} := L^{xy} \sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2} \sqrt{\sigma\tau pq} + \tau L^{xx}. \tag{5.24}$$

Step 2: coercivity of $\mathcal{V}_k$. Apply (5.23) with index $k-1$ and $h = x^k - x$. Then

$$\begin{aligned}\mathcal{V}_k(z) \ge{}& \left(\frac{1}{2} - A_{xy}\right) \left\|x^k - x\right\|^2_{\tau^{-1}p^{-1}} + \frac{1}{2} \left\|y^k - y\right\|^2_{\sigma^{-1}q^{-1}} \\ &+ \left(\frac{1}{p} - 1\right) d_k^2 + \left(\frac{2}{p} - 1\right) r_k^2.\end{aligned}$$

The last two terms are nonnegative. By (5.13), $A_{xy} \le (1-\delta)/2$, and therefore (5.16) follows.

Step 3: bounds for $d^2_{k+1} - r^2_k$, r^2_{k+1}, and Λ_k. By (2.8) and $\bar{x}^{k+1} - x^k = p^{-1}(x^{k+1} - x^k)$,

$$\begin{aligned}\|C_{k+1}\| &\le \frac{L^{xx}}{p} \left\|x^{k+1} - x^k\right\|, \\ \|B_{k+1}\| &\le L^{xy} \left\|y^{k+1} - y^k\right\|.\end{aligned}$$

Substitution into (5.9) yields

$$d^2_{k+1} - r^2_k \le \frac{\tau L^{xx}}{2} \left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} + \frac{L^{xy}}{2} \sqrt{\frac{\sigma\tau p}{q}} \left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}}. \tag{5.25}$$

Moreover,

$$x^{k+1} - \bar{x}^{k+1} = -\frac{1-p}{p}(x^{k+1} - x^k).$$

Using this identity in A_{k+1} and using the already defined B_{k+1} gives

$$r^2_{k+1} \le \frac{(1-p)^2 \tau L^{xx}}{2} \left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}}$$

$$+\frac{L^{xy}}{2}\left(\sqrt{\frac{\sigma\tau p}{q}}-\sqrt{\sigma\tau pq}\right)\left\|y^{k+1}-y^k\right\|^2_{\sigma^{-1}q^{-1}}. \tag{5.26}$$

It remains to estimate Λ_k. Taylor's theorem with Lagrange remainder, applied at (x^k,y^k), gives numbers $\vartheta_{1,k},\vartheta_{2,k},\vartheta_{3,k}\in(0,1)$ such that

$$\begin{aligned}
\Phi(x^{k+1},y^k)&=\Phi(x^k,y^k)+\left\langle\nabla_x\Phi(x^k,y^k),x^{k+1}-x^k\right\rangle\\
&\quad+\frac{1}{2}(x^{k+1}-x^k)^\top\nabla^2_{xx}\Phi\big(x^k+\vartheta_{1,k}(x^{k+1}-x^k),y^k\big)(x^{k+1}-x^k), &(5.27)\\
\Phi(\bar{x}^{k+1},y^k)&=\Phi(x^k,y^k)+\frac{1}{p}\left\langle\nabla_x\Phi(x^k,y^k),x^{k+1}-x^k\right\rangle\\
&\quad+\frac{1}{2p^2}(x^{k+1}-x^k)^\top\nabla^2_{xx}\Phi\left(x^k+\frac{\vartheta_{2,k}}{p}(x^{k+1}-x^k),y^k\right)(x^{k+1}-x^k), &(5.28)\\
\Phi(\bar{x}^{k+1},y^{k+1})&=\Phi(x^k,y^k)+\frac{1}{p}\left\langle\nabla_x\Phi(x^k,y^k),x^{k+1}-x^k\right\rangle+\left\langle\nabla_y\Phi(x^k,y^k),y^{k+1}-y^k\right\rangle\\
&\quad+\frac{1}{2}\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}^\top\nabla^2\Phi\left(\begin{pmatrix}x^k\\ y^k\end{pmatrix}+\vartheta_{3,k}\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}\right)\\
&\qquad\times\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}. &(5.29)
\end{aligned}$$

Substituting (5.27)–(5.29) into (4.7), all first-order terms cancel except the estimator correction, and we obtain

$$\begin{aligned}
\Lambda_k&=\frac{1}{q}\left\langle l^k-\nabla_y\Phi(x^k,y^k),y^{k+1}-y^k\right\rangle\\
&\quad+\frac{1}{2p}(x^{k+1}-x^k)^\top\nabla^2_{xx}\Phi\big(x^k+\vartheta_{1,k}(x^{k+1}-x^k),y^k\big)(x^{k+1}-x^k)\\
&\quad+\frac{q^{-1}-1}{2p^2}(x^{k+1}-x^k)^\top\nabla^2_{xx}\Phi\left(x^k+\frac{\vartheta_{2,k}}{p}(x^{k+1}-x^k),y^k\right)(x^{k+1}-x^k)\\
&\quad-\frac{1}{2q}\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}^\top\nabla^2\Phi\left(\begin{pmatrix}x^k\\ y^k\end{pmatrix}+\vartheta_{3,k}\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}\right)\begin{pmatrix}p^{-1}(x^{k+1}-x^k)\\ y^{k+1}-y^k\end{pmatrix}.
\end{aligned} \tag{5.30}$$

Since

$$\left\|l^k-\nabla_y\Phi(x^k,y^k)\right\|\le\frac{L^{yx}}{p}\left\|x^{k+1}-x^k\right\|,$$

Young's inequality gives

$$\frac{1}{q}\left|\left\langle l^k-\nabla_y\Phi(x^k,y^k),y^{k+1}-y^k\right\rangle\right|\le\frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}}\left(\left\|x^{k+1}-x^k\right\|^2_{\tau^{-1}p^{-1}}+\left\|y^{k+1}-y^k\right\|^2_{\sigma^{-1}q^{-1}}\right). \tag{5.31}$$

Using (2.10), $q^{-1}-1\le q^{-1}$, and

$$\left\|x^{k+1}-x^k\right\|^2=\tau p\left\|x^{k+1}-x^k\right\|^2_{\tau^{-1}p^{-1}},\qquad\left\|y^{k+1}-y^k\right\|^2=\sigma q\left\|y^{k+1}-y^k\right\|^2_{\sigma^{-1}q^{-1}},$$

the three quadratic remainder terms in (5.30) are bounded by

$$\begin{aligned}
&\frac{\tau L^{xx}}{2}\left\|x^{k+1}-x^k\right\|^2_{\tau^{-1}p^{-1}}+\frac{\tau L^{xx}}{2pq}\left\|x^{k+1}-x^k\right\|^2_{\tau^{-1}p^{-1}}\\
&\qquad+\frac{\tau B}{2pq}\left\|x^{k+1}-x^k\right\|^2_{\tau^{-1}p^{-1}}+\frac{\sigma B}{2}\left\|y^{k+1}-y^k\right\|^2_{\sigma^{-1}q^{-1}}.
\end{aligned}$$

Combining this estimate with (5.31) gives

$$|\Lambda_k| \le B_x \left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} + B_y \left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}}, \tag{5.32}$$

where

$$B_x := \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} + \frac{\tau L^{xx}}{2} + \frac{\tau(B + L^{xx})}{2pq}, \tag{5.33}$$

$$B_y := \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} + \frac{\sigma B}{2}. \tag{5.34}$$

Step 4: coercivity of $\widetilde{\mathcal{V}}_k$. Apply (5.23) with indices $k-1$ and k to the two inner products in (5.11). This gives

$$\begin{aligned}\widetilde{\mathcal{V}}_k \ge{}& \frac{1}{2}\left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} + \frac{1}{2}\left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}} - |\Lambda_k| \\ &- \left(\frac{2}{p} - 1\right) A_{xy}\left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} \\ &- \left(\frac{2}{p} - 1\right)\left[(d_{k+1}^2 - r_k^2) + r_{k+1}^2\right].\end{aligned}$$

Indeed, the d_k^2 terms cancel exactly, while the remaining d_{k+1}^2 and r_k^2 terms combine into the last line. Substituting (5.25), (5.26), and (5.32), we obtain

$$\widetilde{\mathcal{V}}_k \ge C_x \left\|x^{k+1} - x^k\right\|^2_{\tau^{-1}p^{-1}} + C_y \left\|y^{k+1} - y^k\right\|^2_{\sigma^{-1}q^{-1}},$$

where

$$\begin{aligned}C_x ={}& \frac{1}{2} - \left(\frac{2}{p} - 1\right)\left[L^{xy}\sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2}\sqrt{\sigma\tau pq} + \frac{3 + (1-p)^2}{2}\tau L^{xx}\right] \\ &- \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} - \frac{\tau L^{xx}}{2} - \frac{\tau(B + L^{xx})}{2pq}, \\ C_y ={}& \frac{1}{2} - \frac{L^{yx}}{2}\sqrt{\frac{\sigma\tau}{pq}} - \frac{\sigma B}{2} \\ &- \left(\frac{2}{p} - 1\right)\left[L^{xy}\sqrt{\frac{\sigma\tau p}{q}} - \frac{L^{xy}}{2}\sqrt{\sigma\tau pq}\right].\end{aligned}$$

Conditions (5.14) and (5.15) imply $C_x, C_y \ge \delta/2$, which proves (5.17). □

Combining (4.6), Lemmas 3, 5, and 6 yields the basic stability inequality.

Theorem 1 (Expected Lyapunov descent)**.** *Under Assumptions 1–3, for every $z \in \mathcal{Z}$,*

$$\mathbb{E}H(\hat{x}^{k+1}, \hat{y}^{k+1}; z) \le \mathbb{E}\mathcal{V}_k(z) - \mathbb{E}\mathcal{V}_{k+1}(z) - \mathbb{E}\widetilde{\mathcal{V}}_k. \tag{5.35}$$

In particular, for every $z^* \in \mathcal{Z}^*$,

$$\mathbb{E}\mathcal{V}_K(z^*) + \sum_{k=0}^{K-1}\mathbb{E}\widetilde{\mathcal{V}}_k \le \mathbb{E}\mathcal{V}_0(z^*) = \frac{1}{2}\mathbb{E}\left\|z^0 - z^*\right\|_V^2, \tag{5.36}$$

$$\sup_{k\ge 0}\mathbb{E}\left\|z^k - z^*\right\|_V^2 \le \frac{1}{\delta}\mathbb{E}\left\|z^0 - z^*\right\|_V^2, \tag{5.37}$$

$$\sum_{k=0}^{\infty}\mathbb{E}\left\|z^{k+1} - z^k\right\|_V^2 \le \frac{1}{\delta}\mathbb{E}\left\|z^0 - z^*\right\|_V^2. \tag{5.38}$$

Proof. The first inequality follows directly from the cited lemmas. For a saddle point z^*, the left-hand gap is nonnegative. Summing from $k = 0$ to $K - 1$ telescopes and proves (5.36). At initialization, the reflected-gradient term and the correction variables vanish, so $\mathbb{E}\mathcal{V}_0(z^*) = \frac{1}{2}\mathbb{E}\left\|z^0 - z^*\right\|_V^2$. The coercivity estimates (5.16) and (5.17) give (5.37) and (5.38). □

6 A realized one-step inequality

The $O(1/K)$ restricted-gap theorem concerns the actual stochastic iterates rather than the virtual full updates. We now compare these two objects. In accordance with the requested notation, define

$$D_g^p(\bar{x}; y) := \left(\frac{1}{p} - 1\right)\big(g(\bar{x}) + \Phi(\bar{x}, y)\big), \tag{6.1}$$

$$D_f^q(x; \bar{y}) := \left(\frac{1}{q} - 1\right)\big(f(\bar{y}) - \Phi(x, \bar{y})\big). \tag{6.2}$$

The superscripts p and q refer to the sampling probabilities; the factors $p^{-1} - 1$ and $q^{-1} - 1$ are built into the definitions.

Define the centered function-value sampling remainders

$$R_g^k := (1 - p^{-1})g(x^k) + p^{-1}g(x^{k+1}) - g(\hat{x}^{k+1}), \tag{6.3}$$

$$R_f^k := (1 - q^{-1})f(y^k) + q^{-1}f(y^{k+1}) - f(\hat{y}^{k+1}). \tag{6.4}$$

By block separability and conditional expectation,

$$\mathbb{E}[R_g^k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[R_f^k \mid \mathcal{F}_{k+\frac{1}{2}}] = 0. \tag{6.5}$$

Set

$$\xi_k := \varepsilon_k + R_g^k + R_f^k. \tag{6.6}$$

By Lemma 4 and (6.5),

$$\mathbb{E}\xi_k = 0. \tag{6.7}$$

The quantities ε_k and ξ_k are scalar and independent of the comparison point (x, y). These two facts are all that will be used when the realized inequality is summed and a supremum over comparison points is taken. Define the comparison-point-dependent martingale field

$$\begin{aligned}\mathfrak{M}_k(x, y) := &\left\langle x, u^k \right\rangle_{\tau^{-1}} + \left\langle y, v^k \right\rangle_{\sigma^{-1}} + \left\langle x, e_x^{k+1} \right\rangle + \left\langle y, e_y^k \right\rangle \\ &- \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle + \left\langle \nabla_y \Phi(x, y^k), v^k \right\rangle.\end{aligned} \tag{6.8}$$

The smooth comparison between actual and virtual coupling values produces the two nonnegative remainder terms

$$\mathscr{R}_k := \frac{\tau L^{xx}}{2}\left\|x^{k+1} - x^k\right\|_{\tau^{-1}p^{-1}}^2 + \frac{\sigma L^{yy}}{2}\left\|y^{k+1} - y^k\right\|_{\sigma^{-1}q^{-1}}^2. \tag{6.9}$$

They will be kept on the right-hand side and controlled by (5.38).

Lemma 7 (Realized one-step gap bound)**.** *Under Assumptions 1–3, for every $k \geq 0$ and every $z = (x, y) \in \mathcal{Z}$,*

$$\begin{aligned}H(x^{k+1}, y^{k+1}; x, y) \leq\ & D_g^p(x^k; y) - D_g^p(x^{k+1}; y) + D_f^q(x; y^k) - D_f^q(x; y^{k+1}) \\ &+ \mathcal{V}_k(z) - \mathcal{V}_{k+1}(z) - \widetilde{\mathcal{V}}_k + \mathscr{R}_k + \mathfrak{M}_k(x, y) + \xi_k.\end{aligned} \tag{6.10}$$

Proof. Start from $H(\hat{x}^{k+1}, \hat{y}^{k+1}; x, y) \le \widehat{\Gamma}_k(z)$ and use Lemma 4 and the exact Lyapunov identity. It remains to compare the actual and virtual gap values.

For the primal part, add and subtract $g(x^k)$ and $\Phi(x^k, y)$. The sampling identity gives

$$g(x^{k+1}) - g(\hat{x}^{k+1}) = (p^{-1} - 1)\big[g(x^k) - g(x^{k+1})\big] + R_g^k.$$

For the coupling, write

$$\begin{aligned}\Phi(x^{k+1}, y) - \Phi(\hat{x}^{k+1}, y) = (p^{-1} - 1)\big[\Phi(x^k, y) - \Phi(x^{k+1}, y)\big] \\ + \Phi(x^k, y) - \Phi(\hat{x}^{k+1}, y) + p^{-1}\big[\Phi(x^{k+1}, y) - \Phi(x^k, y)\big].\end{aligned}$$

The last two terms have conditional mean zero after the linear term is extracted. More directly, L^{xx}-smoothness in the primal variable gives

$$\begin{aligned}&\Phi(x^k, y) - \Phi(\hat{x}^{k+1}, y) + p^{-1}\big[\Phi(x^{k+1}, y) - \Phi(x^k, y)\big] \\ &\quad \le -\left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle + \frac{\tau L^{xx}}{2} \left\| x^{k+1} - x^k \right\|^2_{\tau^{-1}p^{-1}}.\end{aligned}$$

Combining the two displays yields

$$\begin{aligned}&g(x^{k+1}) - g(\hat{x}^{k+1}) + \Phi(x^{k+1}, y) - \Phi(\hat{x}^{k+1}, y) \\ &\quad \le D_g^p(x^k; y) - D_g^p(x^{k+1}; y) - \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle + \frac{\tau L^{xx}}{2} \left\| x^{k+1} - x^k \right\|^2_{\tau^{-1}p^{-1}} + R_g^k.\end{aligned}$$

The dual calculation is analogous, using concavity of $\Phi(x, \cdot)$ or, equivalently, smoothness of $-\Phi(x, \cdot)$:

$$\begin{aligned}&f(y^{k+1}) - f(\hat{y}^{k+1}) + \Phi(x, \hat{y}^{k+1}) - \Phi(x, y^{k+1}) \\ &\quad \le D_f^q(x; y^k) - D_f^q(x; y^{k+1}) + \left\langle \nabla_y \Phi(x, y^k), v^k \right\rangle + \frac{\sigma L^{yy}}{2} \left\| y^{k+1} - y^k \right\|^2_{\sigma^{-1}q^{-1}} + R_f^k.\end{aligned}$$

Adding the two estimates to the virtual gap inequality proves (6.10). □

6.1 Conditional variance and error summability

The next identities quantify precisely how the stochastic errors are controlled by the actual increments.

Lemma 8 (Sampling-error energies)**.** *Let $z^* \in \mathcal{Z}^*$. Then*

$$\sum_{k=0}^{\infty} \mathbb{E}\left\| u^k \right\|^2 \le \frac{\tau(1-p)}{p\delta} \mathbb{E}\left\| z^0 - z^* \right\|_V^2, \tag{6.11}$$

$$\sum_{k=0}^{\infty} \mathbb{E}\left\| v^k \right\|^2 \le \frac{\sigma(1-q)}{q\delta} \mathbb{E}\left\| z^0 - z^* \right\|_V^2, \tag{6.12}$$

$$\sum_{k=0}^{\infty} \mathbb{E}\left\| e_x^{k+1} \right\|^2 \le \frac{\sigma (L^{xy})^2}{q\delta} \mathbb{E}\left\| z^0 - z^* \right\|_V^2, \tag{6.13}$$

$$\sum_{k=0}^{\infty} \mathbb{E}\left\| e_y^{k} \right\|^2 \le \frac{\tau (L^{yx})^2}{p\delta} \mathbb{E}\left\| z^0 - z^* \right\|_V^2. \tag{6.14}$$

Proof. We first treat u^k. By definition,

$$u^k = \hat{x}^{k+1} - x^k - p^{-1}(x^{k+1} - x^k).$$

Consequently,

$$\left\|u^k\right\|_{\tau^{-1}p}^2 = \left\|x^k - \hat{x}^{k+1}\right\|_{\tau^{-1}p}^2 + \left\|x^{k+1} - x^k\right\|_{\tau^{-1}p^{-1}}^2 - 2\left\langle x^k - \hat{x}^{k+1}, x^k - x^{k+1}\right\rangle_{\tau^{-1}}.$$

Since x^k and $\hat{x}^{k+1}$ are $\mathcal{F}_k$-measurable and

$$\mathbb{E}[x^k - x^{k+1} \mid \mathcal{F}_k] = p(x^k - \hat{x}^{k+1}),$$

we obtain

$$\mathbb{E}\left[\left\langle x^k - \hat{x}^{k+1}, x^k - x^{k+1}\right\rangle_{\tau^{-1}} \middle| \mathcal{F}_k\right] = \left\langle x^k - \hat{x}^{k+1}, p(x^k - \hat{x}^{k+1})\right\rangle_{\tau^{-1}} = \left\|x^k - \hat{x}^{k+1}\right\|_{\tau^{-1}p}^2.$$

It follows that

$$\mathbb{E}\left[\left\|u^k\right\|_{\tau^{-1}p}^2 \mid \mathcal{F}_k\right] = \mathbb{E}\left[\left\|x^{k+1} - x^k\right\|_{\tau^{-1}p^{-1}}^2 \middle| \mathcal{F}_k\right] - \left\|x^k - \hat{x}^{k+1}\right\|_{\tau^{-1}p}^2.$$

Using

$$\mathbb{E}[\left\|x^{k+1} - x^k\right\|^2 \mid \mathcal{F}_k] = p\left\|\hat{x}^{k+1} - x^k\right\|^2,$$

we equivalently obtain

$$\mathbb{E}[\left\|u^k\right\|^2 \mid \mathcal{F}_k] = \frac{1-p}{p^2}\mathbb{E}[\left\|x^{k+1} - x^k\right\|^2 \mid \mathcal{F}_k].$$

The same argument, conditionally on $\mathcal{F}_{k+\frac{1}{2}}$, gives

$$\mathbb{E}[\left\|v^k\right\|^2 \mid \mathcal{F}_{k+\frac{1}{2}}] = \frac{1-q}{q^2}\mathbb{E}[\left\|y^{k+1} - y^k\right\|^2 \mid \mathcal{F}_{k+\frac{1}{2}}].$$

Since

$$\left\|x^{k+1} - x^k\right\|^2 = \tau p\left\|x^{k+1} - x^k\right\|_{\tau^{-1}p^{-1}}^2, \qquad \left\|y^{k+1} - y^k\right\|^2 = \sigma q\left\|y^{k+1} - y^k\right\|_{\sigma^{-1}q^{-1}}^2,$$

summation and (5.38) prove (6.11)–(6.12).

For the primal-gradient estimator, set

$$G_x^{k+1} := \nabla_x \Phi(\bar{x}^{k+1}, y^{k+1}) - \nabla_x \Phi(\bar{x}^{k+1}, y^k),$$
$$\widehat{G}_x^{k+1} := \nabla_x \Phi(\bar{x}^{k+1}, \hat{y}^{k+1}) - \nabla_x \Phi(\bar{x}^{k+1}, y^k).$$

By conditional unbiasedness, $\mathbb{E}[G_x^{k+1} \mid \mathcal{F}_{k+\frac{1}{2}}] = q\widehat{G}_x^{k+1}$, and

$$e_x^{k+1} = q^{-1}G_x^{k+1} - \widehat{G}_x^{k+1}.$$

Expanding the square and taking the conditional expectation gives the exact identity

$$\mathbb{E}[\left\|e_x^{k+1}\right\|^2 \mid \mathcal{F}_{k+\frac{1}{2}}] = \frac{1}{q^2}\mathbb{E}[\left\|G_x^{k+1}\right\|^2 \mid \mathcal{F}_{k+\frac{1}{2}}] - \left\|\widehat{G}_x^{k+1}\right\|^2.$$

Therefore, by (2.8),

$$\mathbb{E}\left\|e_x^{k+1}\right\|^2 \le \frac{1}{q^2}\mathbb{E}\left\|G_x^{k+1}\right\|^2$$

$$\leq \frac{(L^{xy})^2}{q^2}\mathbb{E}\left\|y^{k+1}-y^k\right\|^2$$
$$= \frac{\sigma(L^{xy})^2}{q}\mathbb{E}\left\|y^{k+1}-y^k\right\|^2_{\sigma^{-1}q^{-1}}.$$

Summing over k and using (5.38) proves (6.13).

Similarly, with

$$G_y^k := \nabla_y\Phi(x^{k+1},y^k)-\nabla_y\Phi(x^k,y^k),$$
$$\widehat{G}_y^k := \nabla_y\Phi(\hat{x}^{k+1},y^k)-\nabla_y\Phi(x^k,y^k),$$

we have $\mathbb{E}[G_y^k \mid \mathcal{F}_k] = p\widehat{G}_y^k$ and $e_y^k = \widehat{G}_y^k - p^{-1}G_y^k$. Hence

$$\mathbb{E}[\left\|e_y^k\right\|^2 \mid \mathcal{F}_k] = \frac{1}{p^2}\mathbb{E}[\left\|G_y^k\right\|^2 \mid \mathcal{F}_k] - \left\|\widehat{G}_y^k\right\|^2.$$

Using (2.9), summing over k, and applying (5.38) proves (6.14). □

7 Expected $O(1/K)$ restricted-gap convergence

The only comparison-point-dependent sums that require a continuum maximal estimate are

$$y \longmapsto \sum_{k=0}^{K-1}\left\langle \nabla_x\Phi(x^k,y),u^k\right\rangle, \qquad x \longmapsto \sum_{k=0}^{K-1}\left\langle \nabla_y\Phi(x,y^k),v^k\right\rangle.$$

We estimate these two processes directly. We first recall the two analytic facts used in the argument.

Theorem 2 (Sobolev embedding [1])**.** *Let $D\subset\mathbb{R}^d$ be a bounded Lipschitz domain and let $r\in\mathbb{N}$ satisfy $r>d/2$. Then every $h\in W^{r,2}(D)$ has a continuous representative and there is a constant $C_{D,r}>0$ such that*

$$\sup_{z\in D}|h(z)| \leq C_{D,r}\,\|h\|_{W^{r,2}(D)}. \tag{7.1}$$

Theorem 3 (Tonelli's theorem)**.** *If $F:\Omega\times D\to[0,\infty]$ is jointly measurable, then*

$$\mathbb{E}\int_D F(\omega,z)\,\mathrm{d}z = \int_D \mathbb{E}F(\omega,z)\,\mathrm{d}z, \tag{7.2}$$

with both sides allowed to be $+\infty$.

For a multi-index $\alpha=(\alpha_1,\dots,\alpha_N)\in\mathbb{N}_0^N$, write $|\alpha|=\alpha_1+\cdots+\alpha_N$ and

$$D_y^\alpha := \frac{\partial^{|\alpha|}}{\partial y_1^{\alpha_1}\cdots\partial y_N^{\alpha_N}}.$$

When it is applied to the vector field $\nabla_x\Phi$, the derivative is taken componentwise. The notation D_x^β is defined analogously for $\beta\in\mathbb{N}_0^M$.

Let $\mathcal{B}=\mathcal{B}_x\times\mathcal{B}_y\subseteq\mathcal{Z}$ be compact. Since $\mathcal{X}$ and $\mathcal{Y}$ are compact cubes, they are bounded Lipschitz domains. Choose integers

$$\frac{M}{2}<r_x\leq r_0-1, \qquad \frac{N}{2}<r_y\leq r_0-1,$$

and let $C_{\mathcal{X},r_x}$ and $C_{\mathcal{Y},r_y}$ be the corresponding Sobolev embedding constants. For multi-indices α and β, set

$$C_{y,\alpha} := \sup_{(x,y)\in\mathcal{K}}\left\|D_y^\alpha\nabla_x\Phi(x,y)\right\|, \qquad |\alpha|\leq r_y, \tag{7.3}$$

$$C_{x,\beta} := \sup_{(x,y)\in\mathcal{K}} \left\| D_x^\beta \nabla_y \Phi(x,y) \right\|, \qquad |\beta| \le r_x, \tag{7.4}$$

and

$$C'_y := C_{\mathcal{Y},r_y} \left(|\mathcal{Y}| \sum_{|\alpha|\le r_y} C_{y,\alpha}^2 \right)^{1/2}, \tag{7.5}$$

$$C'_x := C_{\mathcal{X},r_x} \left(|\mathcal{X}| \sum_{|\beta|\le r_x} C_{x,\beta}^2 \right)^{1/2}. \tag{7.6}$$

Lemma 9 (Sobolev bounds for the two nonlinear sampling martingales)**.** *For every $K \ge 1$ and every $z^* \in \mathcal{Z}^*$,*

$$\mathbb{E} \sup_{y\in\mathcal{B}_y} \left| \sum_{k=0}^{K-1} \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle \right| \le C'_y \sqrt{\frac{\tau}{p\delta}} \mathbb{E} \left\| z^0 - z^* \right\|_V, \tag{7.7}$$

$$\mathbb{E} \sup_{x\in\mathcal{B}_x} \left| \sum_{k=0}^{K-1} \left\langle \nabla_y \Phi(x, y^k), v^k \right\rangle \right| \le C'_x \sqrt{\frac{\sigma}{q\delta}} \mathbb{E} \left\| z^0 - z^* \right\|_V. \tag{7.8}$$

Proof. We prove (7.7); the second estimate is symmetric. Define the random function

$$S_K(y) := \sum_{k=0}^{K-1} \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle, \qquad y \in \mathcal{Y}.$$

For every multi-index α with $|\alpha| \le r_y$,

$$D_y^\alpha S_K(y) = \sum_{k=0}^{K-1} \left\langle D_y^\alpha \nabla_x \Phi(x^k, y), u^k \right\rangle. \tag{7.9}$$

Because x^k is $\mathcal{F}_k$-measurable and $\mathbb{E}[u^k \mid \mathcal{F}_k] = 0$, the summands in (7.9) are martingale differences for each fixed y. Hence all cross-time terms vanish, and

$$\begin{aligned} \mathbb{E} \left| D_y^\alpha S_K(y) \right|^2 &= \sum_{k=0}^{K-1} \mathbb{E} \left| \left\langle D_y^\alpha \nabla_x \Phi(x^k, y), u^k \right\rangle \right|^2 \\ &\le C_{y,\alpha}^2 \sum_{k=0}^{K-1} \mathbb{E} \left\| u^k \right\|^2 \\ &\le C_{y,\alpha}^2 \frac{\tau}{p\delta} \mathbb{E} \left\| z^0 - z^* \right\|_V^2, \end{aligned}$$

where the last inequality follows from (6.11) and $1 - p \le 1$. Tonelli's theorem gives

$$\mathbb{E} \left\| S_K \right\|_{W^{r_y,2}(\mathcal{Y})}^2 \le |\mathcal{Y}| \left(\sum_{|\alpha|\le r_y} C_{y,\alpha}^2 \right) \frac{\tau}{p\delta} \mathbb{E} \left\| z^0 - z^* \right\|_V^2.$$

Sobolev embedding and Cauchy–Schwarz in probability now yield (7.7) when the initial point and comparison point are deterministic. For an initial-sigma-field measurable pair, condition first on that sigma-field and then use the tower property, which gives the displayed full-expectation form. The proof of (7.8) uses $\mathbb{E}[v^k \mid \mathcal{F}_{k+\frac{1}{2}}] = 0$ and the derivatives $D_x^\beta \nabla_y \Phi$. □

The four linear comparison-point martingales are controlled at the variance level rather than by a radius-times-standard-deviation estimate. Let

$$U_K := \sum_{k=0}^{K-1} u^k, \quad V_K := \sum_{k=0}^{K-1} v^k, \quad E_{x,K} := \sum_{k=0}^{K-1} e_x^{k+1}, \quad E_{y,K} := \sum_{k=0}^{K-1} e_y^k.$$

For every $z = (x, y)$, Young's inequality gives

$$\left|\left\langle x - x^0, U_K \right\rangle_{\tau^{-1}}\right| \le \frac{1}{2}\left\|x - x^0\right\|^2_{\tau^{-1}p^{-1}} + \frac{p}{2\tau}\left\|U_K\right\|^2, \tag{7.10}$$

$$\left|\left\langle y - y^0, V_K \right\rangle_{\sigma^{-1}}\right| \le \frac{1}{2}\left\|y - y^0\right\|^2_{\sigma^{-1}q^{-1}} + \frac{q}{2\sigma}\left\|V_K\right\|^2, \tag{7.11}$$

$$\left|\left\langle x - x^0, E_{x,K} \right\rangle\right| \le \frac{1}{2}\left\|x - x^0\right\|^2_{\tau^{-1}p^{-1}} + \frac{\tau p}{2}\left\|E_{x,K}\right\|^2, \tag{7.12}$$

$$\left|\left\langle y - y^0, E_{y,K} \right\rangle\right| \le \frac{1}{2}\left\|y - y^0\right\|^2_{\sigma^{-1}q^{-1}} + \frac{\sigma q}{2}\left\|E_{y,K}\right\|^2. \tag{7.13}$$

The sequences $u^k, v^k, e_x^{k+1}, e_y^k$ are martingale differences with respect to their natural half-step filtrations. Therefore

$$\mathbb{E}\left\|U_K\right\|^2 = \sum_{k=0}^{K-1} \mathbb{E}\left\|u^k\right\|^2, \qquad \mathbb{E}\left\|V_K\right\|^2 = \sum_{k=0}^{K-1} \mathbb{E}\left\|v^k\right\|^2,$$

$$\mathbb{E}\left\|E_{x,K}\right\|^2 = \sum_{k=0}^{K-1} \mathbb{E}\left\|e_x^{k+1}\right\|^2, \qquad \mathbb{E}\left\|E_{y,K}\right\|^2 = \sum_{k=0}^{K-1} \mathbb{E}\left\|e_y^k\right\|^2.$$

Using Lemma 8, define

$$C_{\mathrm{lin}} := \frac{1-p}{2\delta} + \frac{1-q}{2\delta} + \frac{\tau\sigma p (L^{xy})^2}{2q\delta} + \frac{\tau\sigma q (L^{yx})^2}{2p\delta}. \tag{7.14}$$

Then the variance terms in (7.10)–(7.13) have total expectation at most $C_{\mathrm{lin}}\mathbb{E}\left\|z^0 - z^*\right\|_V^2$. Also set

$$C_{\mathrm{sm}} := \frac{1}{\delta}\max\left\{\frac{\tau L^{xx}}{2}, \frac{\sigma L^{yy}}{2}\right\}. \tag{7.15}$$

By (5.38),

$$\mathbb{E}\sum_{k=0}^{K-1} \mathscr{R}_k \le C_{\mathrm{sm}}\mathbb{E}\left\|z^0 - z^*\right\|_V^2. \tag{7.16}$$

7.1 Restricted-gap theorem

Define

$$\Omega_g^p(\mathcal{B}_y) := \sup_{x\in\mathcal{X},\, y\in\mathcal{B}_y} D_g^p(x;y) - \inf_{x\in\mathcal{X},\, y\in\mathcal{B}_y} D_g^p(x;y), \tag{7.17}$$

$$\Omega_f^q(\mathcal{B}_x) := \sup_{x\in\mathcal{B}_x,\, y\in\mathcal{Y}} D_f^q(x;y) - \inf_{x\in\mathcal{B}_x,\, y\in\mathcal{Y}} D_f^q(x;y). \tag{7.18}$$

Also set

$$R_{\mathcal{B},V}(z^0)^2 := \sup_{z\in\mathcal{B}}\left\|z^0 - z\right\|_V^2 \tag{7.19}$$

and

$$C_{\mathrm{Sob}} := C_y'\sqrt{\frac{\tau}{p\delta}} + C_x'\sqrt{\frac{\sigma}{q\delta}}. \tag{7.20}$$

For $K \ge 1$, define

$$Z^K = (X^K, Y^K) := \frac{1}{K}\sum_{k=0}^{K-1}(x^{k+1}, y^{k+1}). \tag{7.21}$$

Theorem 4 (Expected $O(1/K)$ restricted-gap convergence)**.** *Under Assumptions 1–3, for every $z^* \in \mathcal{Z}^*$ and $K \geq 1$,*

$$\mathbb{E}G_{\mathcal{B}}(Z^K) \leq \frac{C_{\mathcal{B}}(z^0, z^*)}{K}, \tag{7.22}$$

where

$$C_{\mathcal{B}}(z^0, z^*) := \Omega_g^p(\mathcal{B}_y) + \Omega_f^q(\mathcal{B}_x) + \frac{3}{2}\mathbb{E}R_{\mathcal{B},V}(z^0)^2 + (C_{\text{lin}} + C_{\text{sm}})\mathbb{E}\left\|z^0 - z^*\right\|_V^2 + C_{\text{Sob}}\mathbb{E}\left\|z^0 - z^*\right\|_V. \tag{7.23}$$

Proof. Sum (6.10) over $k = 0, \ldots, K-1$. Since $\mathcal{V}_K(z) \geq 0$ and $\widetilde{\mathcal{V}}_k \geq 0$,

$$\sum_{k=0}^{K-1} H(x^{k+1}, y^{k+1}; z) \leq D_g^p(x^0; y) - D_g^p(x^K; y) + D_f^q(x; y^0) - D_f^q(x; y^K) + \frac{1}{2}\left\|z^0 - z\right\|_V^2 + \sum_{k=0}^{K-1} \mathscr{R}_k + \sum_{k=0}^{K-1} \mathfrak{M}_k(z) + \sum_{k=0}^{K-1} \xi_k. \tag{7.24}$$

By convexity of $H(\cdot, z)$,

$$KH(Z^K, z) \leq \sum_{k=0}^{K-1} H(x^{k+1}, y^{k+1}; z).$$

Decompose the four linear parts of $\sum_k \mathfrak{M}_k(z)$ at x^0, y^0. Their base-point terms are independent of z and have zero expectation. The remaining terms satisfy (7.10)–(7.13). Together with the initial Lyapunov term, their comparison-point quadratic contributions total $\frac{3}{2}\left\|z - z^0\right\|_V^2$. The two nonlinear terms are bounded by Lemma 9; the smooth remainder is bounded by (7.16).

The scalar sum $\sum_k \xi_k$ is independent of z. Hence

$$\sup_{z \in \mathcal{B}} \left(F_K(z) + \sum_k \xi_k \right) = \sum_k \xi_k + \sup_{z \in \mathcal{B}} F_K(z),$$

and its expectation vanishes because $\mathbb{E}\xi_k = 0$. Taking the supremum in (7.24), using (7.17)–(7.18), and taking expectations gives

$$K\mathbb{E}G_{\mathcal{B}}(Z^K) \leq C_{\mathcal{B}}(z^0, z^*).$$

This proves the theorem. □

Remark 1 (Why the $O(1/K)$ constant contains a square root)**.** *Only the two nonlinear parameter-indexed martingales require a Sobolev maximal estimate, and that estimate is homogeneous of degree one in their increments. It therefore produces the term $C_{\text{Sob}}\mathbb{E}\left\|z^0 - z^*\right\|_V$ in the full-expectation formulation. The four linear martingale terms are instead controlled by* (7.10)–(7.13) *and contribute at the squared-distance level.*

8 Smoothed gap, restart, and linear convergence

8.1 Smoothed primal–dual gap and quadratic growth

Following the smoothed-gap framework of [6], for $\mu > 0$ define the smoothed primal–dual gap

$$G_\mu(\bar{z}, \dot{z}) := \sup_{z \in \mathcal{Z}} \left\{ H(\bar{z}, z) - \frac{\mu}{2}\left\|z - \dot{z}\right\|_V^2 \right\}. \tag{8.1}$$

The supremum is attained because $\mathcal{Z}$ is compact and the objective is upper semicontinuous in the comparison point. If $z^* \in \mathcal{Z}^*$, then

$$G_\mu(\bar{z}, z^*) \geq H(\bar{z}, z^*) \geq 0. \tag{8.2}$$

Algorithm 2 Restarted nonlinear DSPDHG

Require: outer initial point $Z^0 \in \mathcal{Z}$ and epoch length K

1: **for** $t = 0, 1, 2, \dots$ **do**
2: initialize Algorithm 1 at $z^{t,0} = Z^t$ and reset the history as in (8.8)
3: run K inner iterations and obtain $z^{t,1}, \dots, z^{t,K}$
4: $Z^{t+1} = K^{-1} \sum_{k=1}^{K} z^{t,k}$
5: **end for**

Assumption 4 (Smoothed-gap quadratic growth)**.** *There exist $\mu > 0$ and $\zeta > 0$ such that, for every $\bar{z} \in \mathcal{Z}$ and every $z^* \in \mathcal{Z}^*$,*

$$G_\mu(\bar{z}, z^*) \geq \zeta \operatorname{dist}_V^2(\bar{z}, \mathcal{Z}^*), \qquad \operatorname{dist}_V(\bar{z}, \mathcal{Z}^*) := \inf_{z^* \in \mathcal{Z}^*} \|\bar{z} - z^*\|_V . \tag{8.3}$$

Because every inner iterate belongs to $\mathcal{Z}$ and $\mathcal{Z}$ is convex, every ergodic outer iterate also belongs to $\mathcal{Z}$. Thus no additional assumption that the restarted trajectory remains in a separate compact region is needed.

8.2 Restarted scheme

For epoch $t \geq 0$ and inner iteration $k \in \{0, \dots, K-1\}$, denote the primal and dual samples by $S^{t,k+1}$ and $T^{t,k+1}$. Define the outer-history filtration

$$\mathscr{G}_0 := \{\varnothing, \Omega\}, \qquad \mathscr{G}_t := \sigma\big(S^{r,j}, T^{r,j} : 0 \leq r < t,\ 1 \leq j \leq K\big), \qquad t \geq 1, \tag{8.4}$$

and, within epoch t, define

$$\mathscr{F}_{t,k} := \sigma\big(\mathscr{G}_t, S^{t,1}, T^{t,1}, \dots, S^{t,k}, T^{t,k}\big), \qquad 0 \leq k \leq K, \tag{8.5}$$

$$\mathscr{F}_{t,k+\frac{1}{2}} := \sigma\big(\mathscr{F}_{t,k}, S^{t,k+1}\big), \qquad 0 \leq k \leq K-1. \tag{8.6}$$

Thus $\mathscr{F}_{t,0} = \mathscr{G}_t$ and $\mathscr{G}_{t+1} = \mathscr{F}_{t,K}$. We assume that the samples $S^{t,1}, T^{t,1}, \dots, S^{t,K}, T^{t,K}$ are independent of $\mathscr{G}_t$, mutually independent across inner iterations and half-steps, and have the same laws and marginal probabilities as in (3.1). The outer iterate Z^t is $\mathscr{G}_t$-measurable. For a fixed epoch t, we suppress the epoch index from the inner variables.

Given $Z^t = (X^t, Y^t) \in \mathcal{Z}$, initialize the inner variables at $z^{t,0} = Z^t$, run K inner iterations using the samples above, and set

$$Z^{t+1} := \frac{1}{K} \sum_{k=0}^{K-1} z^{t,k+1}. \tag{8.7}$$

At the beginning of every epoch, reset

$$z^{t,-2} = z^{t,-1} = z^{t,0} = Z^t, \quad w^{t,-1} = w^{t,0} = \nabla_x \Phi(X^t, Y^t), \quad s^{t,0} = w^{t,0}. \tag{8.8}$$

This reset ensures that the initial Lyapunov correction vanishes in every epoch.

Remark 2 (Epochwise validity of the inner analysis)**.** *For each epoch t, replace $\mathcal{F}_k$ and $\mathcal{F}_{k+\frac{1}{2}}$ throughout the inner analysis by $\mathscr{F}_{t,k}$ and $\mathscr{F}_{t,k+\frac{1}{2}}$. Since the epoch samples are independent of $\mathscr{G}_t = \mathscr{F}_{t,0}$ and have the same laws as in the nonrestarted method, all conditional unbiasedness, martingale-difference, Lyapunov, stability, dissipation, and sampling-error arguments remain valid after conditioning on $\mathscr{G}_t$. Taking total expectations and using the tower property gives every preceding expectation estimate, with the same constants, for random $\mathscr{G}_t$-measurable initial points and comparison points; in particular, $\mathbb{E}\varepsilon_k = \mathbb{E}\xi_k = 0$ within each epoch. Hence only total expectations are used in the statements below.*

8.3 A penalized estimate for the two nonlinear martingale fields

For one epoch, define

$$\mathfrak{N}_K(x,y) := -\sum_{k=0}^{K-1} \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle + \sum_{k=0}^{K-1} \left\langle \nabla_y \Phi(x, y^k), v^k \right\rangle. \tag{8.9}$$

Lemma 10 (Penalized nonlinear martingale estimate)**.** *Let $z^* = (x^*, y^*) \in \mathcal{Z}^*$ be deterministic or measurable with respect to the epoch initial sigma-field. Then*

$$\begin{aligned} \mathbb{E} \sup_{z=(x,y)\in\mathcal{Z}} \left\{ \mathfrak{N}_K(x,y) - \frac{\mu K}{4} \|z - z^*\|_V^2 \right\} \\ \le \frac{C_{\mathrm{nl}}}{\mu} \mathbb{E} \left\| z^0 - z^* \right\|_V^2, \end{aligned} \tag{8.10}$$

where

$$C_{\mathrm{nl}} := \frac{\tau\sigma}{\delta} \left[\frac{q(1-p)}{p} (L^{xy})^2 + \frac{p(1-q)}{q} (L^{yx})^2 \right]. \tag{8.11}$$

Proof. Split the first field at y^*:

$$\begin{aligned} -\sum_k \left\langle \nabla_x \Phi(x^k, y), u^k \right\rangle = & -\sum_k \left\langle \nabla_x \Phi(x^k, y^*), u^k \right\rangle \\ & -\sum_k \left\langle \nabla_x \Phi(x^k, y) - \nabla_x \Phi(x^k, y^*), u^k \right\rangle. \end{aligned}$$

The first sum is independent of y and has conditional expectation zero. By (2.8),

$$\begin{aligned} \left| \sum_k \left\langle \nabla_x \Phi(x^k, y) - \nabla_x \Phi(x^k, y^*), u^k \right\rangle \right| &\le L^{xy} \|y - y^*\| \sum_k \left\| u^k \right\| \\ &= L^{xy} \sqrt{\sigma q} \, \|y - y^*\|_{\sigma^{-1}q^{-1}} \sum_k \left\| u^k \right\| \\ &\le \frac{\mu K}{4} \|y - y^*\|_{\sigma^{-1}q^{-1}}^2 + \frac{\sigma q (L^{xy})^2}{\mu K} \left(\sum_k \left\| u^k \right\| \right)^2. \end{aligned}$$

Similarly, splitting the second field at x^* gives a base-point martingale of conditional mean zero and

$$\left| \sum_k \left\langle \nabla_y \Phi(x, y^k) - \nabla_y \Phi(x^*, y^k), v^k \right\rangle \right| \le \frac{\mu K}{4} \|x - x^*\|_{\tau^{-1}p^{-1}}^2 + \frac{\tau p (L^{yx})^2}{\mu K} \left(\sum_k \left\| v^k \right\| \right)^2.$$

Cauchy–Schwarz in time gives $(\sum_k \|u^k\|)^2 \le K \sum_k \|u^k\|^2$ and the analogous inequality for v^k. Taking total expectations, using the conditional martingale-difference identities and the tower property, and applying (6.11)–(6.12) proves (8.10). □

8.4 Endpoint correction estimate

Let

$$a_p := p^{-1} - 1, \qquad a_q := q^{-1} - 1, \qquad \alpha := a_p + a_q. \tag{8.12}$$

For an epoch starting from z^0 and ending at z^K, define

$$\mathscr{E}_K(z) := D_g^p(x^0; y) - D_g^p(x^K; y) + D_f^q(x; y^0) - D_f^q(x; y^K). \tag{8.13}$$

Lemma 11 (Endpoint corrections)**.** *Let $z_1^* = (x_1^*, y_1^*) \in \mathcal{Z}^*$. For every $z = (x, y) \in \mathcal{Z}$,*

$$\begin{aligned}\mathscr{E}_K(z) \le \alpha H(z^0, z_1^*) &+ a_p \ell_{xy} \left\|x^0 - x^K\right\|_{\tau^{-1}p^{-1}} \|y - y_1^*\|_{\sigma^{-1}q^{-1}} \\ &+ a_q \ell_{xy} \|x - x_1^*\|_{\tau^{-1}p^{-1}} \left\|y^K - y^0\right\|_{\sigma^{-1}q^{-1}}. \end{aligned} \tag{8.14}$$

Consequently, for every $z_2^* \in \mathcal{Z}^*$,

$$\mathscr{E}_K(z) \le \alpha H(z^0, z_1^*) + \|z - z_2^*\|_V^2 + \|z_1^* - z_2^*\|_V^2 + c_D \left\|z^0 - z^K\right\|_V^2, \tag{8.15}$$

where

$$c_D := \frac{\ell_{xy}^2}{2} \max\{a_p^2, a_q^2\}. \tag{8.16}$$

Proof. For the primal correction, insert y_1^*:

$$\begin{aligned} &g(x^0) + \Phi(x^0, y) - g(x^K) - \Phi(x^K, y) \\ &= g(x^0) + \Phi(x^0, y_1^*) - g(x^K) - \Phi(x^K, y_1^*) \\ &\quad + \Phi(x^0, y) - \Phi(x^K, y) - \Phi(x^0, y_1^*) + \Phi(x^K, y_1^*). \end{aligned}$$

Since x_1^* minimizes $L(\cdot, y_1^*)$ and y_1^* maximizes $L(x_1^*, \cdot)$,

$$\begin{aligned} &g(x^0) + \Phi(x^0, y_1^*) - g(x^K) - \Phi(x^K, y_1^*) \\ &= L(x^0, y_1^*) - L(x^K, y_1^*) \\ &\le L(x^0, y_1^*) - L(x_1^*, y_1^*) \\ &\le L(x^0, y_1^*) - L(x_1^*, y^0) = H(z^0, z_1^*). \end{aligned}$$

For $s, t \in [0, 1]$, set

$$x_s := x^K + s(x^0 - x^K), \qquad y_t := y_1^* + t(y - y_1^*).$$

The fundamental theorem of calculus in both variables gives

$$\begin{aligned} &\Phi(x^0, y) - \Phi(x^K, y) - \Phi(x^0, y_1^*) + \Phi(x^K, y_1^*) \\ &= \int_0^1 \int_0^1 \left\langle \nabla_{xy}^2 \Phi(x_s, y_t)(y - y_1^*), x^0 - x^K \right\rangle \mathrm{d}t \, \mathrm{d}s. \end{aligned}$$

By (2.11), its absolute value is at most

$$\ell_{xy} \left\|x^0 - x^K\right\|_{\tau^{-1}p^{-1}} \|y - y_1^*\|_{\sigma^{-1}q^{-1}}.$$

The dual correction is identical after inserting x_1^* and using that y_1^* maximizes $L(x_1^*, \cdot)$. This proves (8.14).

Using the fixed Young inequality $ab \le (a^2 + b^2)/2$,

$$\begin{aligned} a_p \ell_{xy} \left\|x^0 - x^K\right\|_{\tau^{-1}p^{-1}} \|y - y_1^*\|_{\sigma^{-1}q^{-1}} &\le \frac{1}{2} \|y - y_1^*\|_{\sigma^{-1}q^{-1}}^2 + \frac{a_p^2 \ell_{xy}^2}{2} \left\|x^0 - x^K\right\|_{\tau^{-1}p^{-1}}^2, \\ a_q \ell_{xy} \|x - x_1^*\|_{\tau^{-1}p^{-1}} \left\|y^K - y^0\right\|_{\sigma^{-1}q^{-1}} &\le \frac{1}{2} \|x - x_1^*\|_{\tau^{-1}p^{-1}}^2 + \frac{a_q^2 \ell_{xy}^2}{2} \left\|y^K - y^0\right\|_{\sigma^{-1}q^{-1}}^2. \end{aligned}$$

Therefore the comparison-point contribution is at most $\frac{1}{2}\|z - z_1^*\|_V^2$, which is bounded by $\|z - z_2^*\|_V^2 + \|z_1^* - z_2^*\|_V^2$. The endpoint-increment terms are bounded by $c_D \left\|z^0 - z^K\right\|_V^2$. This proves (8.15). □

8.5 One-epoch estimate

For an epoch initialized at z^0, let

$$Z^K := \frac{1}{K}\sum_{k=1}^{K} z^k$$

be its ergodic output.

Lemma 12 (One-epoch smoothed-gap estimate)**.** *Suppose Assumptions 1–3 hold and*

$$K \geq \frac{16}{\mu}. \tag{8.17}$$

Then, for every $z_1^*, z_2^* \in \mathcal{Z}^*$ that are deterministic or measurable with respect to the epoch initial sigma-field,

$$\mathbb{E}G_\mu(Z^K, z_2^*) \leq \frac{\alpha}{K}\mathbb{E}G_\mu(z^0, z_1^*) + \frac{1}{K}\mathbb{E}\left\|z_1^* - z_2^*\right\|_V^2 + \frac{C_\mu}{K}\mathbb{E}\left\|z^0 - z_2^*\right\|_V^2, \tag{8.18}$$

where

$$C_\mu := 3 + C_{\mathrm{lin}} + C_{\mathrm{sm}} + 2\left(1 + \frac{1}{\delta}\right)c_D + \frac{C_{\mathrm{nl}}}{\mu}. \tag{8.19}$$

Proof. Summing (6.10), using convexity of $H(\cdot, z)$, and dropping the nonnegative final Lyapunov and dissipation terms gives

$$KH(Z^K, z) \leq \mathscr{E}_K(z) + \frac{1}{2}\left\|z^0 - z\right\|_V^2 + \sum_{k=0}^{K-1}\mathscr{R}_k + \mathfrak{L}_K(z) + \mathfrak{N}_K(z) + \sum_{k=0}^{K-1}\xi_k, \tag{8.20}$$

where $\mathfrak{N}_K$ is defined in (8.9) and

$$\mathfrak{L}_K(z) := \langle x, U_K\rangle_{\tau^{-1}} + \langle y, V_K\rangle_{\sigma^{-1}} + \langle x, E_{x,K}\rangle + \langle y, E_{y,K}\rangle.$$

Decompose $\mathfrak{L}_K(z)$ at z^0. Its base-point part is independent of z and has zero total expectation by conditional centering and the tower property. By (7.10)–(7.13),

$$\mathfrak{L}_K(z) \leq \left\|z - z^0\right\|_V^2 + \mathfrak{B}_K + \text{a centered, } z\text{-independent term}, \tag{8.21}$$

where

$$\mathfrak{B}_K := \frac{p}{2\tau}\left\|U_K\right\|^2 + \frac{q}{2\sigma}\left\|V_K\right\|^2 + \frac{\tau p}{2}\left\|E_{x,K}\right\|^2 + \frac{\sigma q}{2}\left\|E_{y,K}\right\|^2,$$

and, by Lemma 8,

$$\mathbb{E}\mathfrak{B}_K \leq C_{\mathrm{lin}}\mathbb{E}\left\|z^0 - z_2^*\right\|_V^2.$$

Moreover, (7.16) applies with the random epoch-initial comparison point $z^* = z_2^*$ by Remark 2.

Subtract $\frac{\mu K}{2}\left\|z - z_2^*\right\|_V^2$ from (8.20). Use Lemma 11 and split the smoothing penalty into two equal halves. Lemma 10 controls $\mathfrak{N}_K$ together with the first half. For the remaining deterministic comparison-point terms,

$$\begin{aligned}\frac{1}{2}\left\|z^0 - z\right\|_V^2 + \left\|z - z^0\right\|_V^2 &= \frac{3}{2}\left\|z - z^0\right\|_V^2 \\ &\leq 3\left\|z - z_2^*\right\|_V^2 + 3\left\|z^0 - z_2^*\right\|_V^2.\end{aligned}$$

The endpoint estimate contributes one further copy of $\left\|z - z_2^*\right\|_V^2$. Hence all remaining positive comparison-point quadratics are absorbed by $-\frac{\mu K}{4}\left\|z - z_2^*\right\|_V^2$ whenever $\mu K/4 \geq 4$, which is (8.17).

Now take the supremum over $z \in \mathcal{Z}$ and then total expectations. The scalar sum $\sum_k \xi_k$ and all base-point linear martingales are independent of z. By Remark 2, their total expectations vanish after applying the appropriate conditional martingale-difference identities and the tower property. We obtain

$$K\mathbb{E}G_\mu(Z^K, z_2^*) \le \alpha\mathbb{E}H(z^0, z_1^*) + \mathbb{E}\left\|z_1^* - z_2^*\right\|_V^2 + \left(3 + C_{\mathrm{lin}} + C_{\mathrm{sm}} + \frac{C_{\mathrm{nl}}}{\mu}\right)\mathbb{E}\left\|z^0 - z_2^*\right\|_V^2 + c_D\mathbb{E}\left\|z^0 - z^K\right\|_V^2.$$

By (5.37),

$$\mathbb{E}\left\|z^0 - z^K\right\|_V^2 \le 2\mathbb{E}\left\|z^0 - z_2^*\right\|_V^2 + 2\mathbb{E}\left\|z^K - z_2^*\right\|_V^2 \le 2\left(1 + \frac{1}{\delta}\right)\mathbb{E}\left\|z^0 - z_2^*\right\|_V^2.$$

Finally, $H(z^0, z_1^*) \le G_\mu(z^0, z_1^*)$ pointwise. Division by K proves (8.18). □

8.6 Outer recurrence and linear convergence

The saddle-point set $\mathcal{Z}^*$ is a nonempty compact subset of $\mathcal{Z}$. Indeed, the primal value function $x \mapsto \sup_{y\in\mathcal{Y}} L(x,y)$ is lower semicontinuous on the compact set $\mathcal{X}$, whereas the dual value function $y \mapsto \inf_{x\in\mathcal{X}} L(x,y)$ is upper semicontinuous on the compact set $\mathcal{Y}$; since a saddle point exists, the corresponding primal and dual solution sets are nonempty compact sets and their Cartesian product equals $\mathcal{Z}^*$. Hence the V-metric projection multifunction onto $\mathcal{Z}^*$ has nonempty compact values and a closed graph. By the measurable selection theorem, it admits a Borel measurable selection. Therefore, for each $t \ge 0$, we may choose an $\mathscr{G}_t$-measurable projection

$$\Pi^t \in \operatorname*{arg\,min}_{z^*\in\mathcal{Z}^*} \left\|Z^t - z^*\right\|_V, \tag{8.22}$$

and define

$$a_t := \mathbb{E}G_\mu(Z^t, \Pi^{t-1}), \qquad t \ge 1. \tag{8.23}$$

Lemma 13 (Movement of consecutive projections)**.** *For every $t \ge 2$,*

$$\mathbb{E}\left\|\Pi^t - \Pi^{t-1}\right\|_V^2 \le \frac{2}{\zeta}a_t + \frac{2}{\delta\zeta}a_{t-1}. \tag{8.24}$$

Proof. By the squared triangle inequality,

$$\left\|\Pi^t - \Pi^{t-1}\right\|_V^2 \le 2\left\|Z^t - \Pi^t\right\|_V^2 + 2\left\|Z^t - \Pi^{t-1}\right\|_V^2.$$

Quadratic growth gives $\mathbb{E}\left\|Z^t - \Pi^t\right\|_V^2 \le \zeta^{-1}a_t$. Since Z^t is the average of the inner iterates of epoch $t-1$, Jensen's inequality and (5.37) give

$$\begin{aligned}\mathbb{E}\left\|Z^t - \Pi^{t-1}\right\|_V^2 &\le \frac{1}{K}\sum_{k=1}^K \mathbb{E}\left\|z^{t-1,k} - \Pi^{t-1}\right\|_V^2 \\ &\le \frac{1}{\delta}\mathbb{E}\left\|Z^{t-1} - \Pi^{t-1}\right\|_V^2 \le \frac{1}{\delta\zeta}a_{t-1}.\end{aligned}$$

Combining the two bounds proves the claim. □

Applying Lemma 12 to epoch t, with $z_1^* = \Pi^{t-1}$ and $z_2^* = \Pi^t$, and then using Lemma 13 and quadratic growth, yields for $t \ge 2$

$$a_{t+1} \le \frac{\alpha\zeta + 2 + C_\mu}{K\zeta}a_t + \frac{2}{K\delta\zeta}a_{t-1}. \tag{8.25}$$

Theorem 5 (Linear convergence of restarted nonlinear DSPDHG)**.** *Suppose Assumptions 1–4 hold and choose*

$$K \geq \max\left\{\frac{16}{\mu}, \frac{100}{\zeta}(\alpha\zeta + 2 + C_\mu), \frac{200}{\delta\zeta}\right\}. \tag{8.26}$$

Then there exists $M_0 < \infty$ such that

$$\mathbb{E}G_\mu(Z^t, \Pi^{t-1}) \leq M_0 5^{-t}, \qquad t \geq 1, \tag{8.27}$$

$$\mathbb{E}\,\mathrm{dist}_V^2(Z^t, \mathcal{Z}^*) \leq \frac{M_0}{\zeta} 5^{-t}, \qquad t \geq 1. \tag{8.28}$$

Proof. Condition (8.26) reduces (8.25) to

$$a_{t+1} \leq \frac{1}{100}(a_t + a_{t-1}), \qquad t \geq 2. \tag{8.29}$$

Let $r = 1/5$ and choose

$$M_0 \geq \max\{5a_1, 25a_2\}.$$

Then $a_1 \leq M_0 r$ and $a_2 \leq M_0 r^2$. Suppose $a_{t-1} \leq M_0 r^{t-1}$ and $a_t \leq M_0 r^t$. By (8.29),

$$a_{t+1} \leq \frac{M_0}{100}(r^t + r^{t-1}) = M_0 r^{t-1}\frac{1+r}{100} \leq M_0 r^{t+1},$$

because $r^2 = 1/25 > (1+r)/100$. Induction proves (8.27). Finally, since $\Pi^{t-1} \in \mathcal{Z}^*$, quadratic growth gives

$$\zeta\,\mathrm{dist}_V^2(Z^t, \mathcal{Z}^*) \leq G_\mu(Z^t, \Pi^{t-1}),$$

and taking expectations proves (8.28). □

9 Conclusion

The nonlinear doubly stochastic PDHG method has an expected ergodic restricted-gap rate $O(1/K)$ under compact-domain local smoothness and the explicit Lyapunov step-size conditions. Under smoothed-gap quadratic growth, its fixed-period restarted version satisfies a 5^{-t} linear rate in both the expected smoothed gap and the expected squared distance to the saddle-point set. The proof separates the four linear martingale terms, which are controlled directly at the variance level, from the two nonlinear parameter-indexed martingales, which require Sobolev control for the sublinear theorem and quadratic-penalty absorption for the restart theorem.

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